5 Stereochemistry

Goals for Chapter 5

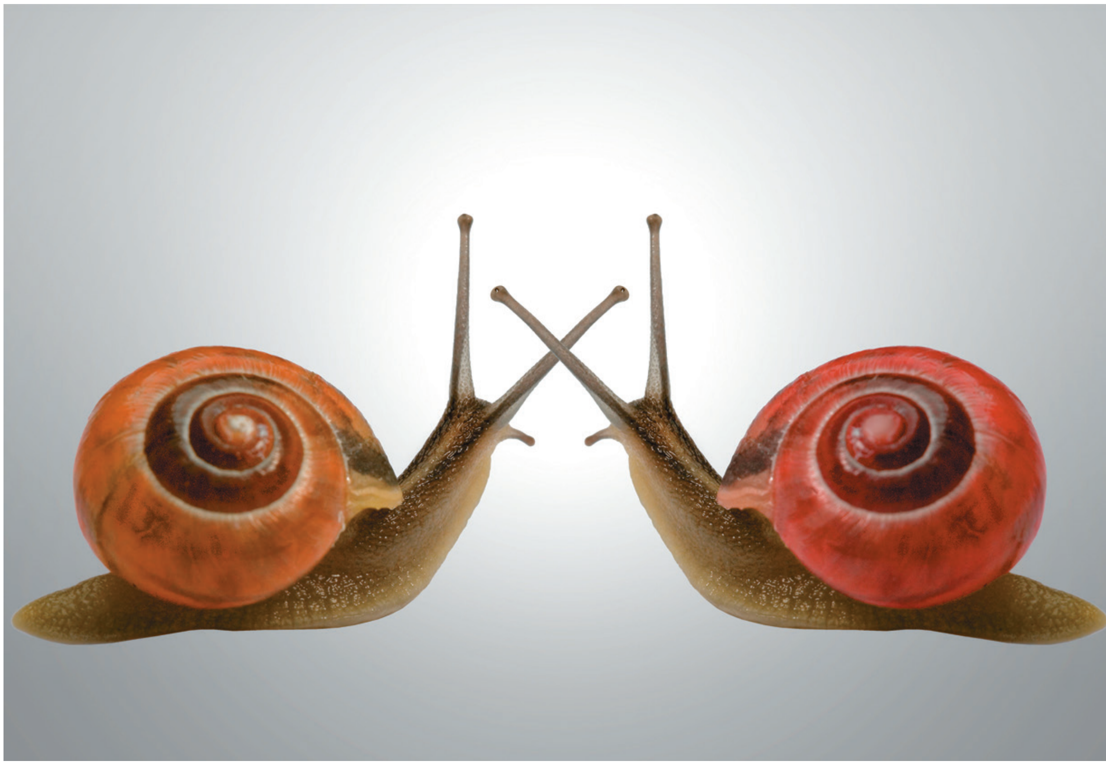
1 Recognize structures that have stereoisomers, and identify the relationships between the stereoisomers.

2 Recognize chiral structures, draw their mirror images, and identify features that may suggest chirality.

3 Identify asymmetric carbon atoms and other stereocenters, and assign their configurations.

4 Explain the relationships between optical activity and chirality, optical purity, and enantiomeric excess.

5 Explain how the different types of stereoisomers differ in their physical and chemical properties.

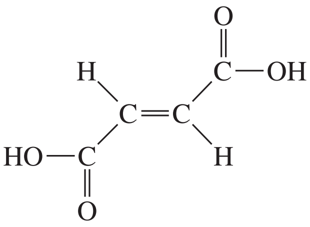


▸ Chirality is not an obscure scientific concept. Almost all of nature is chiral, and all living things are chiral. Hands, feet, paws, plants, flowers, hooves, protein, DNA, and people are all chiral. This photo shows two otherwise identical snails with mirror-image shells (Section 5-2).

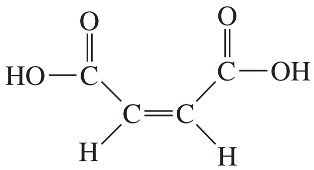
5-1 Introduction

Stereochemistry is the study of the three-dimensional structure of molecules. No one can understand organic chemistry, biochemistry, or biology without using stereochemistry. Biological systems are exquisitely selective, and they often discriminate between molecules with subtle stereochemical differences. We have seen (Section 1-19) that isomers are grouped into two broad classes: constitutional isomers and stereoisomers. **Constitutional isomers** (**structural isomers**) differ in their bonding sequence; their atoms are connected differently. **Stereoisomers** (also called **configurational isomers**) have the same bonding sequence, but they differ in the orientation of their atoms in space.

Differences in spatial orientation might seem unimportant, but stereoisomers often have remarkably different physical, chemical, and biological properties. For example, the cis and trans isomers of butenedioic acid (next page) are a special type of stereoisomers called *cis-trans isomers* (or *geometric isomers*). Both compounds have the formula HOOC—CH═CH—COOH, but they differ in how these atoms are arranged in space. The cis isomer is called *maleic acid*, and the trans isomer is called *fumaric acid*. Fumaric acid is an essential metabolic intermediate in both plants and animals, but maleic acid is toxic and irritating to tissues.



fumaric acid, mp 287 °C
essential metabolite



maleic acid, mp 138 °C
toxic irritant

The discovery of stereochemistry was one of the most important breakthroughs in the structural theory of organic chemistry. Stereochemistry explained why several types of isomers exist, and it forced scientists to propose the tetrahedral carbon atom. In this chapter, we study the three-dimensional structures of molecules to understand their stereochemical relationships. We compare the various types of stereoisomers and study ways to differentiate among stereoisomers. In later chapters, we will see how stereochemistry plays a major role in the properties and reactions of organic compounds.

5-2 Chirality

What is the difference between your left hand and your right hand? They look similar, yet a left-handed glove does not fit the right hand. The same principle applies to your feet. They look almost identical, yet the left shoe fits painfully on the right foot. The relationship between your two hands or your two feet is that they are nonsuperimposable (nonidentical) mirror images of each other. Objects that have left-handed and right-handed forms are called **chiral** (*kī′rel*, rhymes with "spiral"), the Greek word for "handed."

We can tell whether an object is chiral by looking at its mirror image (Figure 5-1). Every physical object (with the possible exception of a vampire) has a mirror image, but *a chiral object has a mirror image that is different from the original object*. For example, a chair and a spoon and a glass of water all look the same in a mirror. Such objects are called **achiral,** meaning "not chiral." A hand looks different in the mirror. If the original hand were the right hand, it would look like a left hand in the mirror.

Application: Mirror-Image People

Most people have their heart on their left side and their liver and appendix on the right. About 1 person in 10,000 is born with a congenital condition called *situs inversus,* in which their major organs are in the mirror image positions, with the heart on the right and the liver and appendix on the left. People with situs inversus are approximately (but not in minute detail) the mirror image of the majority of people.

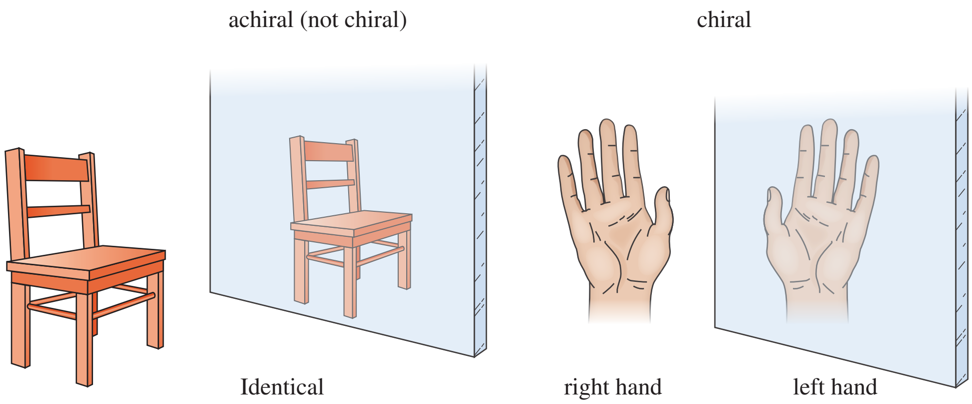


FIGURE 5-1 Use of a mirror to test for chirality. An object is chiral if its mirror image is different from the original object.

Besides shoes and gloves, we encounter many other chiral objects every day (Figure 5-2). What is the difference between an English car and an American car? The English car has the steering wheel on the right-hand side, while the American car has it on the left. To a first approximation, the English and American cars are nonsuperimposable mirror images. Most screws have right-hand threads and are turned clockwise to tighten. The mirror image of a right-handed screw is a left-handed screw, turned counterclockwise to tighten. Those of us who are left-handed realize that scissors are chiral. Most scissors are right-handed. If you use them in your left hand, they cut poorly, if at all. A left-handed person must go to a well-stocked store (or a specialized "Leftorium") to find a pair of left-handed scissors, the mirror image of the "standard" right-handed scissors.

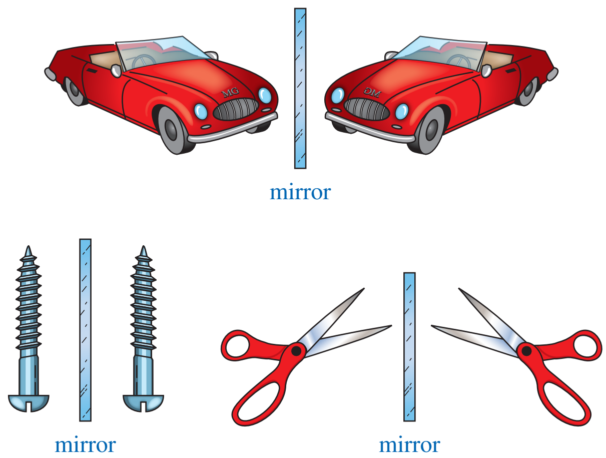
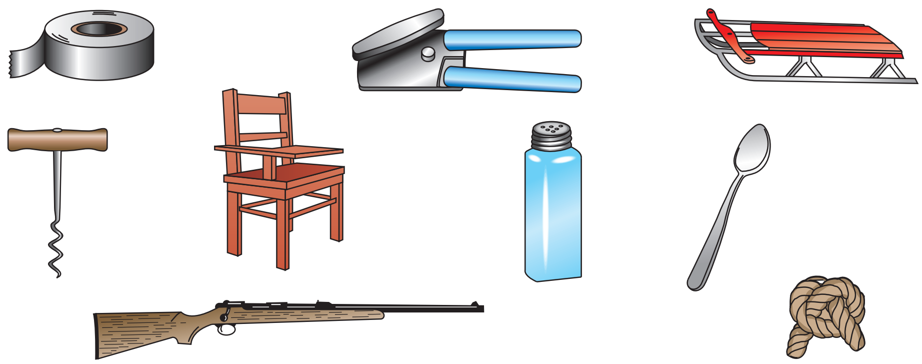


FIGURE 5-2 Common chiral objects. Many objects come in "left-handed" and "right-handed" versions.

PROBLEM 5-1

Determine whether the following objects are chiral or achiral.



5-2A Chirality and Enantiomers in Organic Molecules

Like other objects, molecules are either chiral or achiral. For example, consider the two geometric isomers of 1,2-dichlorocyclopentane (Figure 5-3). The cis isomer is achiral because its mirror image is superimposable on the original molecule. Two molecules are said to be **superimposable** if they can be placed on top of each other and the three-dimensional position of each atom of one molecule coincides with the equivalent atom of the other molecule. To draw the mirror image of a molecule, simply draw the same structure with left and right reversed. The up-and-down and front-and-back directions are unchanged. The two mirror-image structures of the cis isomer are identical (superimposable), and *cis*-1,2-dichlorocyclopentane is achiral.

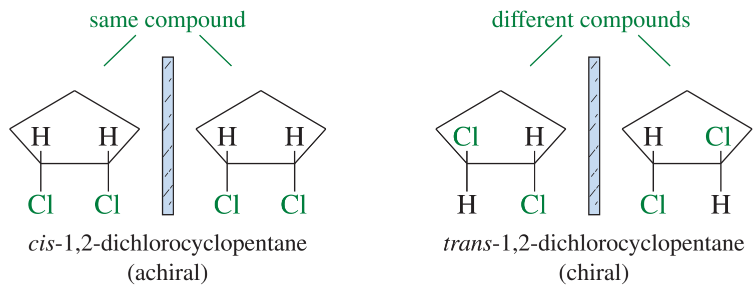


FIGURE 5-3 Stereoisomers of 1,2-dichlorocyclopentane. The cis isomer has no enantiomers; it is achiral. The trans isomer is chiral; it can exist in either of two nonsuperimposable enantiomeric forms.

The mirror image of *trans*-1,2-dichlorocyclopentane is different from (nonsuperimposable with) the original molecule. These are two different compounds, and we should expect to discover two mirror-image isomers of *trans*-1,2-dichlorocyclopentane. Make models of these isomers to convince yourself that they are different no matter how you twist and turn them. Nonsuperimposable mirror-image molecules are called **enantiomers.** A chiral compound always has an enantiomer (a nonsuperimposable mirror image). An achiral compound always has a mirror image that is the same as the original molecule. Let's review the definitions of these words.

PROBLEM-SOLVING HINT

Every object has a mirror image. Is its mirror image the same or different? Different: The object is chiral. Same: The object is achiral.

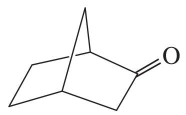
enantiomers:	mirror-image isomers; pairs of compounds that are nonsuperimposable mirror images
chiral:	("handed") different from its mirror image; having an enantiomer
achiral:	("not handed") identical with its mirror image; not chiral

Any compound that is chiral must have an enantiomer. Any compound that is achiral cannot have an enantiomer.

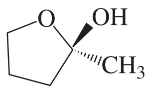
PROBLEM 5-2

Make a model and draw a three-dimensional structure for each compound. Then draw the mirror image of your original structure and determine whether the mirror image is the same compound. Label each structure as being chiral or achiral, and label pairs of enantiomers.

(a) *cis*-1,2-dimethylcyclobutane
(b) *trans*-1,2-dimethylcyclobutane
(c) *cis*- and *trans*-1,3-dimethylcyclobutane
(d) 2-bromobutane
(e)



(f)



PROBLEM-SOLVING HINT

Stereochemistry is a difficult topic for many students. Use your models to help you see the relationships between structures. Once you have experience working with these three-dimensional relationships, you may be able to visualize them without constructing models.

5-2B Asymmetric Carbon Atoms, Chiral Centers, and Stereocenters

The three-dimensional drawing of 2-bromobutane in Figure 5-4 shows that 2-bromobutane cannot be superimposed on its mirror image. This simple molecule is chiral, with two distinct enantiomers. What is it about a molecule that makes it chiral? The most common feature (but not the only one) that leads to chirality is a carbon atom that is bonded to four different groups. Such a carbon atom is called an **asymmetric carbon atom** or a **chiral carbon atom,** and is often designated by an asterisk (*). Carbon atom 2 of 2-bromobutane

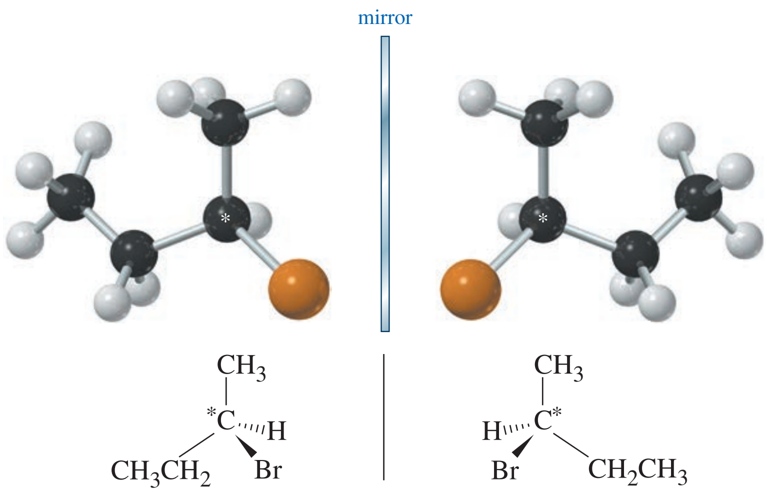


FIGURE 5-4 Enantiomers (nonsuperimposable mirror image isomers) of 2-bromobutane. 2-Bromobutane is chiral by virtue of an asymmetric carbon atom (chiral carbon atom), marked with an *.

is bonded to a hydrogen atom, a bromine atom, a methyl group, and an ethyl group. It is an asymmetric carbon atom, and it is responsible for the chirality of 2-bromobutane.

An asymmetric carbon atom is the most common example of a **chiral center** (or **chirality center**), the IUPAC term for any atom holding a set of ligands in a spatial arrangement that is not superimposable on its mirror image. Chiral centers belong to an even broader group called *stereocenters*. A **stereocenter** (or **stereogenic atom**) is any atom at which the interchange of two groups gives a stereoisomer.* Asymmetric carbons and the double-bonded carbon atoms in cis-trans isomers are the most common types of stereocenters. Figure 5-5 compares these successively broader definitions.

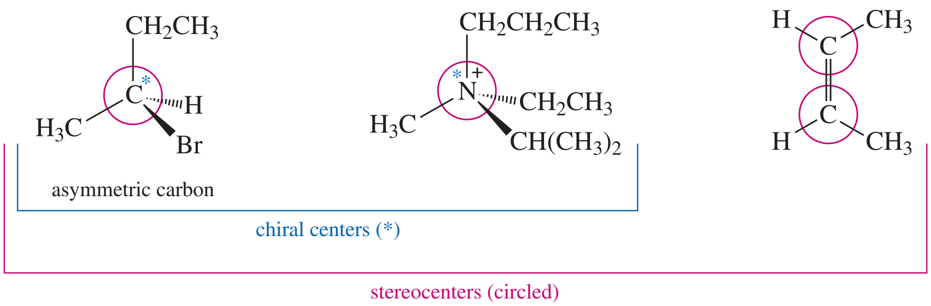


FIGURE 5-5 Asymmetric carbon atoms are examples of chiral centers, which are examples of stereocenters.

Make a model of an asymmetric carbon atom, bonded to four different-colored atoms. Also make its mirror image and try to superimpose the two (Figure 5-6). No matter how you twist and turn the models, they never look exactly the same.

If two of the four groups on a carbon atom are the same, however, the arrangement usually is not chiral. Figure 5-7 shows the mirror image of a tetrahedral structure with only three different groups; two of the four groups are the same. If the structure on the right is rotated 180°, it can be superimposed on the left structure.

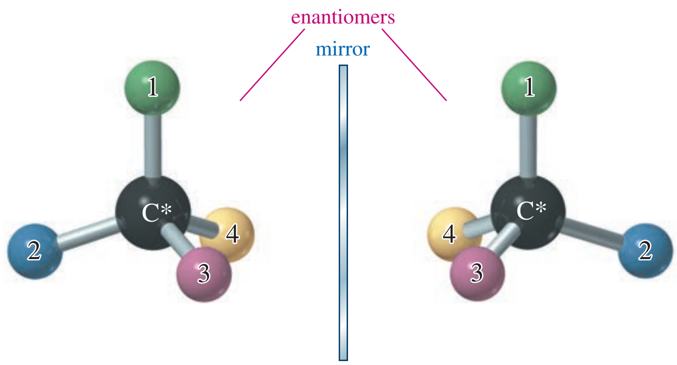


FIGURE 5-6 Enantiomers of an asymmetric carbon atom. These two mirror images are nonsuperimposable.

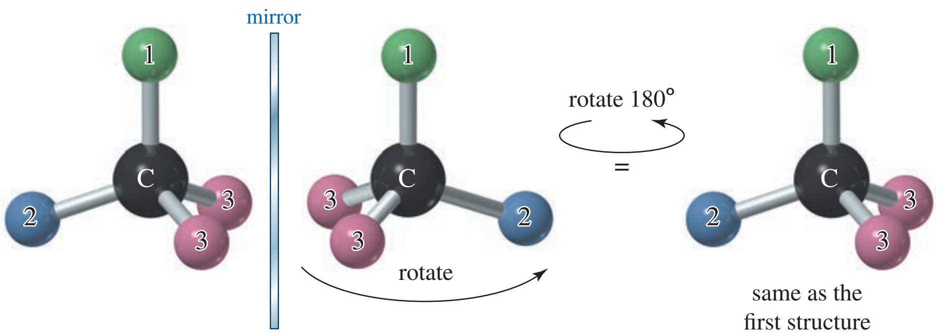


FIGURE 5-7 A carbon atom bonded to just three different types of groups is not chiral.

*The term *stereocenter* (*stereogenic atom*) is not consistently defined. The original (Mislow) definition is given here. Some sources simply define it as a synonym for an *asymmetric carbon* (*chiral carbon*) or for a *chiral center*.

We can generalize at this point, but keep in mind that the ultimate test for chirality is always whether the molecule's mirror image is the same or different.

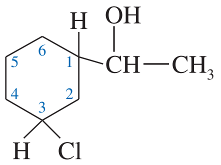
1. If a compound has no asymmetric carbon atom, it is usually achiral. (We will see exceptions in Section 5-9.)
2. If a compound has just one asymmetric carbon atom, it must be chiral.
3. If a compound has more than one asymmetric carbon, it may or may not be chiral. (We will see examples in Section 5-12.)

COMMON TERMINOLOGY

Many organic chemists use the broad term *stereocenter* to mean "chiral center." You may need to determine the meaning from the context. Most of the rules that apply to asymmetric carbons and other chiral centers do not apply to the stereocenters in double bonds.

SOLVED PROBLEM 5-1

Identify each asymmetric carbon atom in the following structure:



SOLUTION

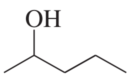
This structure contains three asymmetric carbons:

1. The (CHOH) carbon of the side chain is asymmetric. Its four substituents are the ring, a hydrogen atom, a hydroxy group, and a methyl group.
2. Carbon atom C1 of the ring is asymmetric. Its four substituents are the side chain, a hydrogen atom, the part of the ring closer to the chlorine atom ($-CH_2-CHCl-$), and the part of the ring farther from the chlorine atom ($-CH_2-CH_2-CH_2-CHCl-$).
3. The ring carbon C3 bearing the chlorine atom is asymmetric. Its four substituents are the chlorine atom, a hydrogen atom, the part of the ring closer to the side chain, and the part of the ring farther from the side chain.

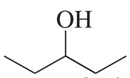
Notice that different groups might be different in any manner. For example, the ring carbon bearing the chlorine atom is asymmetric even though two of its ring substituents initially appear to be $-CH_2-$ groups. These two parts of the ring are different because one is closer to the side chain and one is farther away. The *entire* structure of the group must be considered.

PROBLEM 5-3

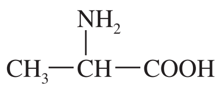
Draw a three-dimensional structure for each compound, and star all asymmetric carbon atoms. Draw the mirror image for each structure, and state whether you have drawn a pair of enantiomers or just the same molecule twice. Build molecular models of any of these examples that seem difficult to you.

(a)


pentan-2-ol

(b)


pentan-3-ol

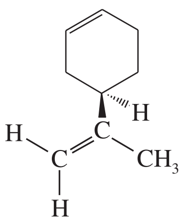
(c)


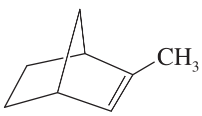
alanine

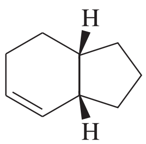
(d) 1-bromo-2-methylbutane

(e) chlorocyclohexane

(f) *cis*-1,2-dichlorocyclobutane

(g)


(h)


(i)


PROBLEM-SOLVING HINT

To draw the mirror image of a structure, keep up-and-down and front-and-back aspects as they are in the original structure, but reverse left and right.

PROBLEM-SOLVING HINT

To determine whether a ring carbon is asymmetric, see if there is a difference in the path around the ring in each direction. If there is, then the two ring bonds are "different groups."

PROBLEM 5-4

For each of the stereocenters (circled) in Figure 5-5,
(a) draw the compound with two of the groups on the stereocenter interchanged.
(b) give the relationship of the new compound to the original compound.

5-2C Mirror Planes of Symmetry

In Figure 5-3, we saw that *cis*-1,2-dichlorocyclopentane is achiral. Its mirror image was found to be identical with the original molecule. Figure 5-8 illustrates a shortcut that often shows whether a molecule is chiral. If we draw a line down the middle of *cis*-1,2-dichlorocyclopentane, bisecting a carbon atom and its two hydrogen atoms, the part of the molecule that appears to the right of the line is the mirror image of the part on the left. This kind of symmetry is called an **internal mirror plane,** sometimes symbolized by the Greek lowercase letter sigma (σ). Because the right-hand side of the molecule is the reflection of the left-hand side, the molecule's mirror image is the same as the original molecule.

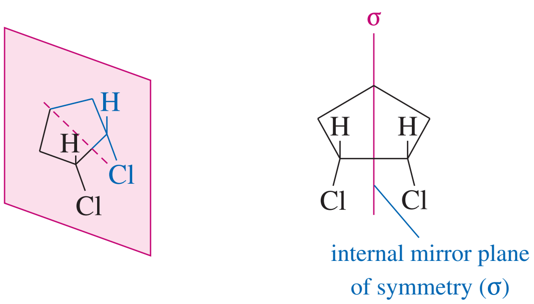
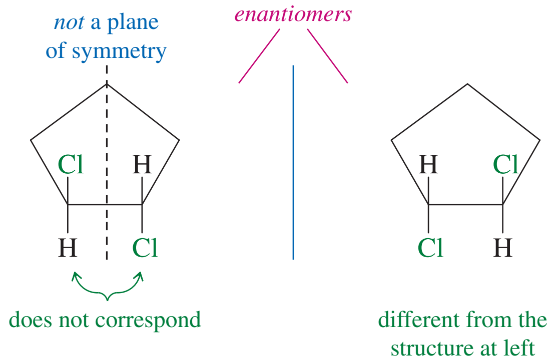


FIGURE 5-8 Internal mirror plane. *cis*-1,2-Dichlorocyclopentane has an internal mirror plane of symmetry. Any compound with an internal mirror plane of symmetry *cannot* be chiral.

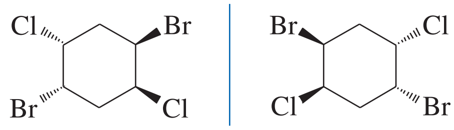
Notice in the following figure that the chiral trans isomer of 1,2-dichlorocyclopentane does not have a mirror plane of symmetry. The chlorine atoms do not reflect into each other across our hypothetical mirror plane. One of them is directed up, and the other is directed down.



We can generalize from these and other examples to state the following principle:

> *Any molecule that has an internal mirror plane of symmetry cannot be chiral*, even though it may contain asymmetric carbon atoms.

The converse is not true, however. When we cannot find a mirror plane of symmetry, that does not necessarily mean that the molecule must be chiral. The following example has no internal mirror plane of symmetry, yet the mirror image is superimposable on the original molecule. You may need to make models to show that these mirror images are just two drawings of the same compound.



(identical mirror images)

Using what we know about mirror planes of symmetry, we can see why a chiral (asymmetric) carbon atom is special. Figure 5-4 showed that an asymmetric carbon has a mirror image that is nonsuperimposable on the original structure; it has no internal mirror plane of symmetry. If a carbon atom has only three different kinds of substituents, however, it has an internal mirror plane of symmetry (Figure 5-9). Therefore, it cannot contribute to chirality in a molecule.

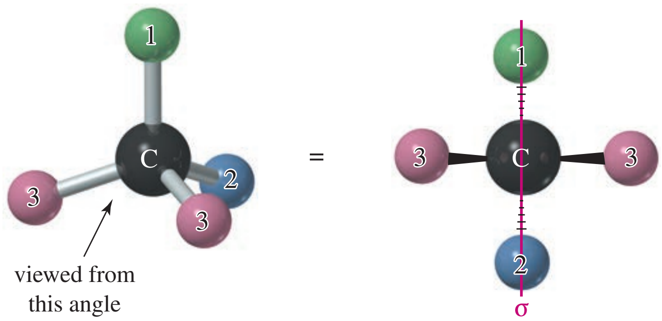
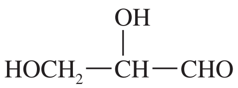


FIGURE 5-9 A carbon atom with two identical substituents (only three different substituents) usually has an internal mirror plane of symmetry. The structure is not chiral.

PROBLEM 5-5

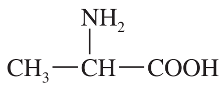
For each compound, determine whether the molecule has an internal mirror plane of symmetry. If it does, draw the mirror plane on a three-dimensional drawing of the molecule. If the molecule does not have an internal mirror plane, determine whether the structure is chiral.

(a) methane
(b) *cis*-1,2-dibromocyclobutane
(c) *trans*-1,2-dibromocyclobutane
(d) 1,2-dichloropropane
(e)



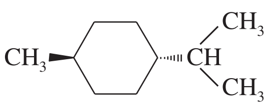
glyceraldehyde

(f)



alanine

(g)



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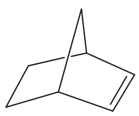
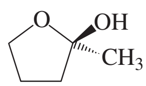
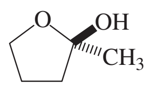


PROBLEM-SOLVING HINT

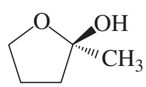
Various methods are used to show the stereochemistry of bonds. Bonds toward the viewer are shown as heavy or dark, often with perspective (the end toward the viewer appears larger). Bonds away from the viewer are shown as dashed or dotted, with or without perspective. The IUPAC has recently recommended the convention that we follow in this book. All four structures shown below depict exactly the same molecule.



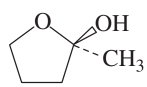
perspective



no perspective



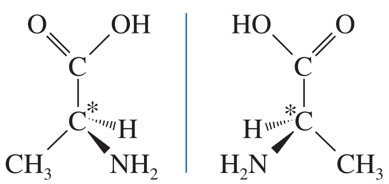
IUPAC convention



other options

5-3 (*R*) and (*S*) Nomenclature of Asymmetric Carbon Atoms

Alanine, from Problem 5-5(f), is one of the amino acids found in common proteins. Alanine has an asymmetric carbon atom, and it exists in two enantiomeric forms.



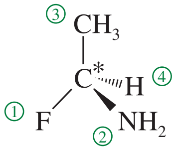
natural alanine unnatural alanine

These mirror images are different, and this difference is reflected in their biochemistry. Only the enantiomer on the left can be metabolized by the usual enzyme; the one on the right is not recognized as a useful amino acid. Both are named alanine, however, or 2-aminopropanoic acid in the IUPAC system. We need a simple way to distinguish between enantiomers and to give each of them a unique name.

The difference between the two enantiomers of alanine lies in the three-dimensional arrangement of the four groups around the asymmetric carbon atom. Any asymmetric carbon has two possible (mirror-image) spatial arrangements, which we call **configurations.** The alanine enantiomers represent the two possible arrangements of its four groups around the asymmetric carbon atom. If we can name the two configurations of any asymmetric carbon atom, then we have a way of specifying and naming the enantiomers of alanine or any other chiral compound.

The **Cahn–Ingold–Prelog convention** is the most widely accepted system for naming the configurations of chiral centers. Each asymmetric carbon atom is assigned a letter (R) or (S) based on its three-dimensional configuration. To determine the name, we follow a two-step procedure that assigns "priorities" to the four substituents and then assigns the name based on the relative positions of these substituents. Here is the procedure:

1. *Assign a relative "priority" to each group bonded to the asymmetric carbon.* We speak of group 1 as having the highest priority, group 2 as having the second priority, group 3 as having the third priority, and group 4 as having the lowest priority.

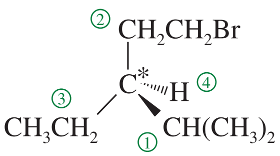


(a) Look at the first atom of the group—the atom bonded to the asymmetric carbon. *Atoms with higher atomic numbers receive higher priorities.* For example, consider the structure shown at right. If the four groups bonded to an asymmetric carbon atom are H, CH_3, NH_2, and F, the fluorine atom (atomic number 9) receives the highest priority, followed by the nitrogen atom of the NH_2 group (atomic number 7), then by the carbon atom of the methyl group (atomic number 6). Note that we look only at the atomic number of the atom directly attached to the asymmetric carbon, not the entire group. Hydrogen would have the lowest priority.

With different isotopes of the same element, the heavier isotopes have higher priorities. For example, tritium (^3H) receives a higher priority than deuterium (^2H), followed by hydrogen (^1H).

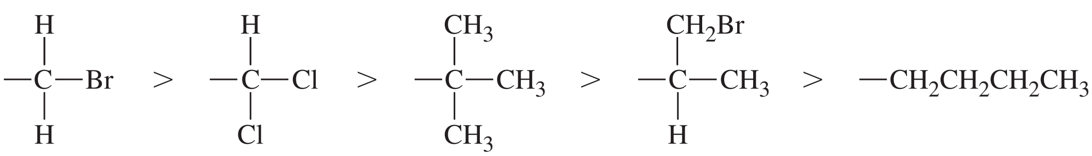
Examples of priority for atoms bonded to an asymmetric carbon:

$$I > Br > Cl > S > F > O > N > {}^{13}C > {}^{12}C > Li > {}^{3}H > {}^{2}H > {}^{1}H$$

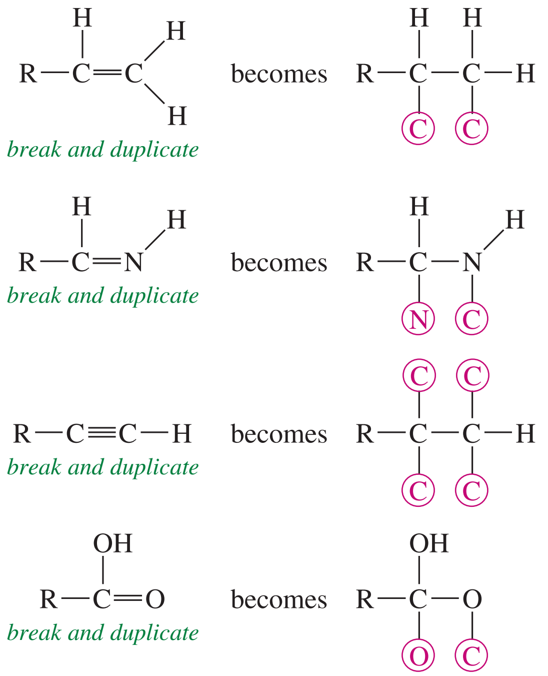


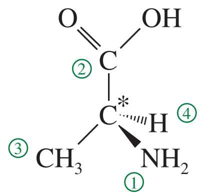
(b) *In case of ties, use the next atoms along the chain of each group as tiebreakers.* For example, in the structure at right, we assign a higher priority to isopropyl $—CH(CH_3)_2$ than to ethyl $—CH_2CH_3$ or bromoethyl $—CH_2CH_2Br$. The first carbon in the isopropyl group is bonded to two carbons, while the first carbon in the ethyl group (or the bromoethyl group) is bonded to only one carbon. An ethyl group and a $—CH_2CH_2Br$ have identical first atoms and second atoms, but the bromine atom in the third position gives $—CH_2CH_2Br$ a higher priority than $—CH_2CH_3$. One high-priority atom takes priority over any number of lower-priority atoms.

Examples

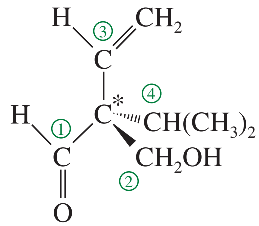


(c) *Treat double and triple bonds as if each were a bond to a separate atom*. For this method, imagine that each pi bond is broken and the atoms at both ends duplicated. Note that when you break a bond, you always add *two* imaginary atoms. (Imaginary atoms are circled below.) Examples of assigning priorities with double bonds are shown at left.





alanine



2. *Using a three-dimensional drawing or a model, put the fourth-priority group away from you and view the molecule with the first, second, and third priority groups radiating toward you like the spokes of a steering wheel* (Figure 5-10). *Draw an arrow from the first-priority group, through the second, to the third.* If the arrow points clockwise, the asymmetric carbon atom is called (*R*) (Latin, *rectus*, "upright"). If the arrow points counterclockwise, the chiral carbon atom is called (*S*) (Latin, *sinister*, "left").

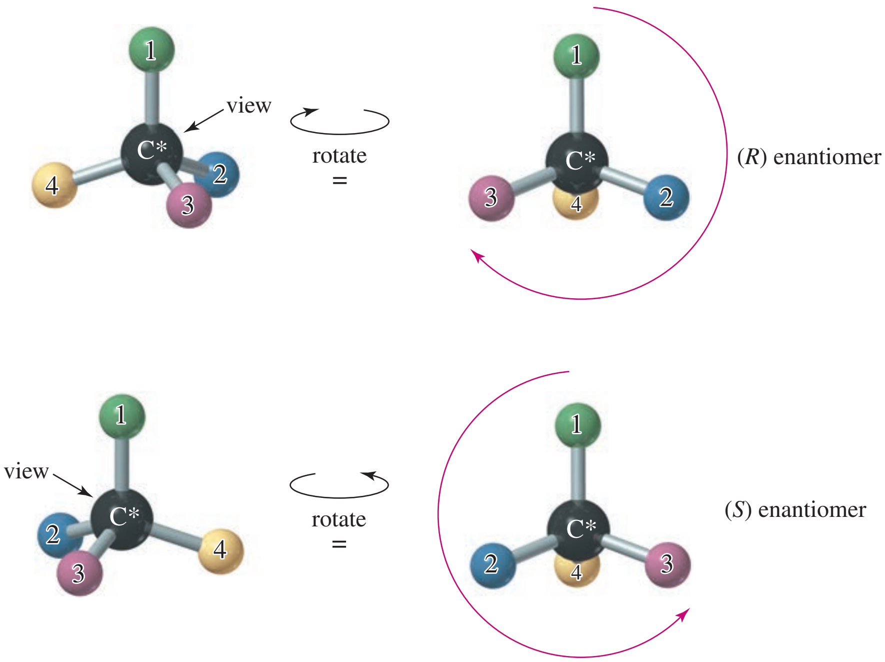


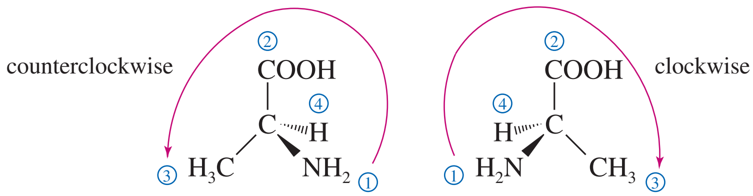
FIGURE 5-10 Assignment of (*R*) and (*S*) labels to asymmetric carbon atoms. View the molecule with the lowest- (fourth-) priority atom (often hydrogen) away from you. Then draw a circular arrow from group 1 to group 2 to group 3. A clockwise arrow (to the right) corresponds to the (*R*) configuration, and a counterclockwise arrow (to the left) corresponds to (*S*).

Alternatively, you can draw the arrow and imagine turning a car's steering wheel in that direction. If the car would go to the left, the asymmetric carbon atom is designated (*S*). If the car would go to the right, the asymmetric carbon atom is designated (*R*).

Let's use the enantiomers of alanine as an example. The naturally occurring enantiomer is the one on the left, determined to have the (*S*) configuration. Of the four atoms attached to the asymmetric carbon in alanine, nitrogen has the largest atomic number, giving it the highest priority. Next is the —COOH carbon atom, because it is bonded to oxygen atoms. Third is the methyl group, followed by the hydrogen atom. When we position the natural enantiomer with its hydrogen atom pointing away from us, the arrow from $-NH_2$ to —COOH to $-CH_3$ points counterclockwise. Thus, the naturally occurring enantiomer of alanine has the (*S*) configuration. Make models of these enantiomers to illustrate how they are named (*R*) and (*S*).

PROBLEM-SOLVING HINT

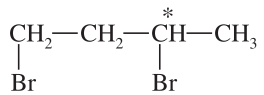
Until you become comfortable working with drawings, use models to help you assign (*R*) and (*S*) configurations.



natural (*S*)-alanine   unnatural (*R*)-alanine

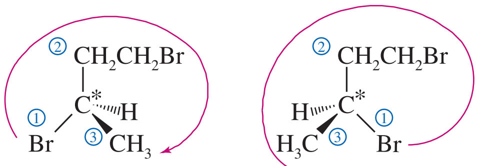
SOLVED PROBLEM 5-2

Draw the enantiomers of 1,3-dibromobutane and label them as (*R*) and (*S*). (Making a model is particularly helpful for this type of problem.)



SOLUTION

The third carbon atom in 1,3-dibromobutane is asymmetric. The bromine atom receives first priority, the ($-CH_2CH_2Br$) group second priority, the methyl group third, and the hydrogen fourth. The following mirror images are drawn with the hydrogen atom back, ready to assign (*R*) or (*S*) as shown.



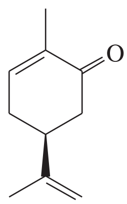
(*R*)   (*S*)

PROBLEM-SOLVING HINT

In assigning priorities for a ring carbon, go around the ring in each direction until you find a point of difference; then use the difference to determine which ring carbon has higher priority than the other.

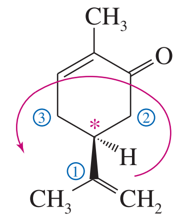
SOLVED PROBLEM 5-3

The structure of one of the enantiomers of carvone is shown here. Find the asymmetric carbon atom, and determine whether it has the (*R*) or the (*S*) configuration.

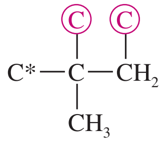


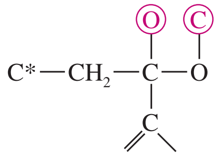
SOLUTION

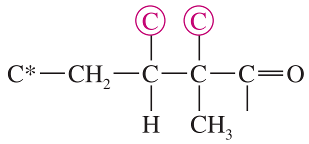
The asymmetric carbon atom is one of the ring carbons, as indicated by the asterisk in the following structure. Although two $—CH_2—$ groups are bonded to the carbon, they are different $—CH_2—$ groups. One is a $—CH_2—CO—$ group, and the other is a $—CH_2—CH{=}C$ group. The groups are assigned priorities, and this is found to be the (*S*) enantiomer.



(*S*)-carvone

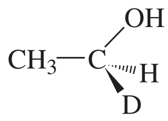
Group 1: 

Group 2: 

Group 3: 

Application: Biochemistry

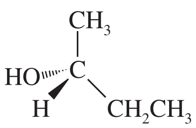
Scientists frequently use the isotopes of hydrogen to assign the configuration of the products of biological reactions. Ethanol, made chiral by the presence of a deuterium (D or ^{2}H), is one of the early examples.

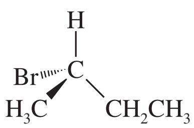


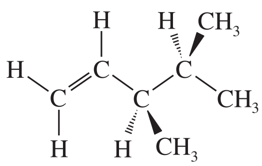
(*S*)-1-deuterioethanol

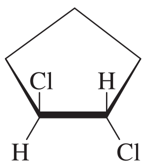
PROBLEM 5-6

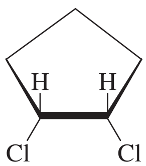
Star (*) each asymmetric carbon atom in the following examples, and determine whether it has the (*R*) or (*S*) configuration.

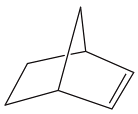
(a) 

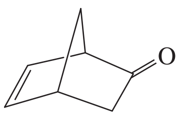
(b) 

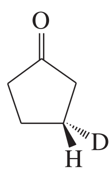
(c) 

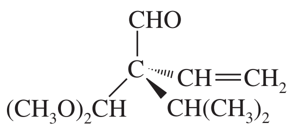
(d) 

(e) 

(f) 

(g) 

(h) 

(*i) 

PROBLEM-SOLVING HINT

If the lowest-priority atom (usually H) is oriented toward you, you don't need to turn the structure around. You can leave it as it is with the H toward you, find the (*R*) or (*S*) configuration, and reverse your answer.

PROBLEM 5-7

In Problem 5-3, you drew the enantiomers for a number of chiral compounds. Now go back and designate each asymmetric carbon atom as either (*R*) or (*S*).

PROBLEM-SOLVING HINT

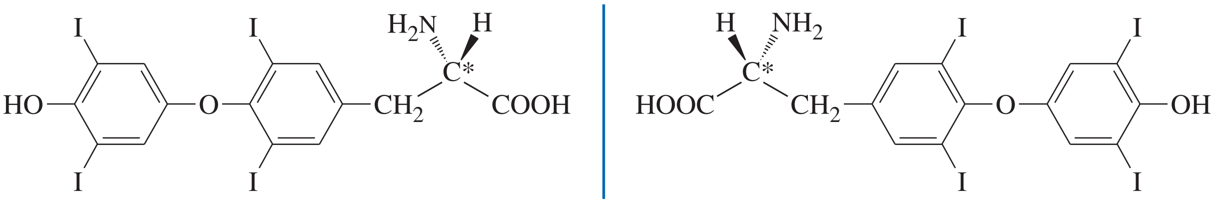
Interchanging any two substituents on an asymmetric carbon atom inverts its (*R*) or (*S*) configuration. If there is only one chiral center in a molecule, inverting its configuration gives the enantiomer.

5-4 Optical Activity

Mirror-image molecules have identical physical properties in an achiral environment. Compare the following properties of (*R*)-2-bromobutane and (*S*)-2-bromobutane.

	(*R*)-2-Bromobutane	**(*S*)-2-Bromobutane**
boiling point (°C)	91.2	91.2
melting point (°C)	−112	−112
refractive index	1.436	1.436
density	1.253	1.253

Differences in enantiomers become apparent in their interactions with other chiral molecules, such as enzymes. Still, we need a simple method to distinguish between enantiomers and measure their purity in the laboratory. **Polarimetry** is a common method used to distinguish between enantiomers, based on their ability to rotate the plane of polarized light in opposite directions. For example, the two enantiomers of thyroid hormone are shown below. The (*S*) enantiomer has a powerful effect on the metabolic rate of all the cells in the body. The (*R*) enantiomer is useless. In the laboratory, we distinguish between the enantiomers by observing that the active one rotates the plane of polarized light to the left.



thyroid hormone (*S*)
rotates polarized light to the left

wrong enantiomer (*R*)
rotates polarized light to the right

5-4A Plane-Polarized Light

Most of what we see is unpolarized light, consisting of waves vibrating randomly in all directions. **Plane-polarized light** is composed of waves that vibrate in only one plane. Although there are other types of "polarized light," the term usually refers to plane-polarized light.

When unpolarized light passes through a polarizing filter, the randomly vibrating light waves are filtered so that most of the light passing through is vibrating in one direction (Figure 5-11). The direction of vibration is called the *axis* of the filter. Polarizing filters may be made from carefully cut calcite crystals or from specially treated plastic sheets. Plastic polarizing filters are often used as lenses in sunglasses, because the axis of the filters can be positioned to filter out reflected glare.

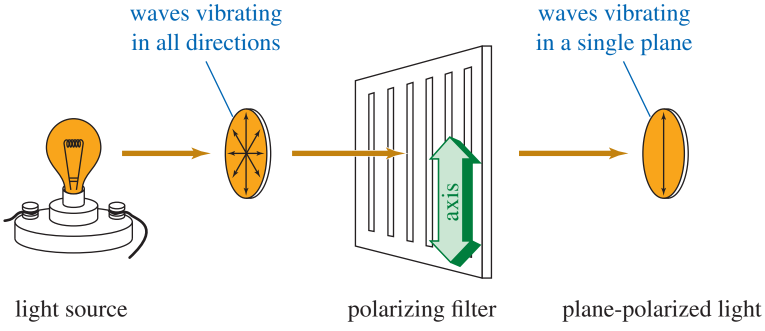


FIGURE 5-11 Function of a polarizing filter. The waves of plane-polarized light vibrate primarily in a single plane.

When light passes first through one polarizing filter and then through another, the amount of light emerging depends on the relative orientation of the axes of the two filters (Figure 5-12). If the axes of the two filters are lined up (parallel), then nearly all the light that passes through the first filter also passes through the second. If the axes of the two filters are perpendicular (*crossed poles*), however, all the polarized light that emerges from the first filter is stopped by the second. At intermediate angles of rotation, intermediate amounts of light pass through.

You can demonstrate this effect for yourself by wearing a pair of polarized sunglasses while looking at a light source through another pair (Figure 5-13). The second pair seems to be transparent, as long as its axis is lined up with the pair you are wearing. When the second pair is rotated to 90°, however, the lenses become opaque, as if they were covered with black ink.

5-4B Rotation of Plane-Polarized Light

When polarized light passes through a solution containing a chiral compound, the chiral compound causes the plane of vibration to rotate. Rotation of the plane of polarized light is called **optical activity,** and substances that rotate the plane of polarized light are said to be **optically active.**

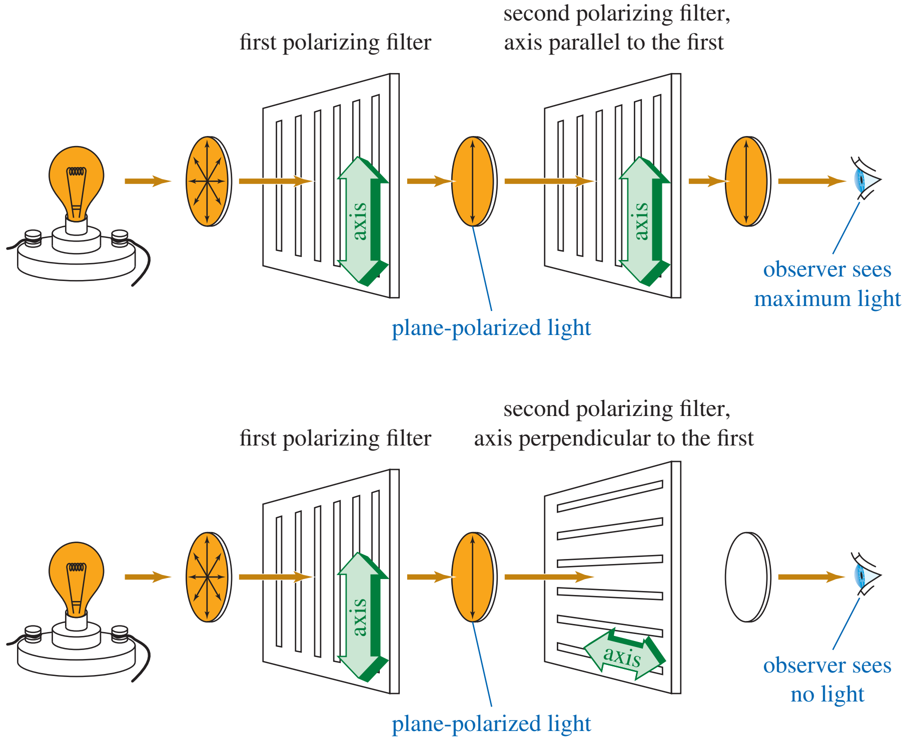


FIGURE 5-12 Crossed poles. When the axis of a second polarizing filter is parallel to the first, a maximum amount of light passes through. When the axes of the filters are perpendicular (crossed poles), no light passes through.

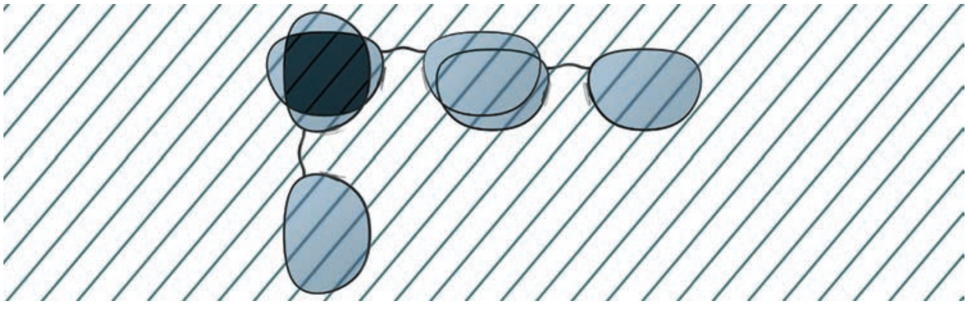


FIGURE 5-13 Using sunglasses to demonstrate parallel axes of polarization and crossed poles. When the pairs of sunglasses are parallel, a maximum amount of light passes through. When they are perpendicular, very little light passes through.

Before the relationship between chirality and optical activity was known, enantiomers were called **optical isomers** because they seemed identical except for their opposite optical activity. The term was loosely applied to more than one type of isomerism among optically active compounds, however, and this ambiguous term has been replaced by the well-defined term *enantiomers*.

Two enantiomers have identical physical properties, except for the direction they rotate the plane of polarized light.

> Enantiomeric compounds rotate the plane of polarized light by exactly the same amount but in opposite directions.

PROBLEM-SOLVING HINT

Don't confuse the process for *naming* a structure (*R*) or (*S*) with the process for *measuring* an optical rotation. Just because we use the terms *clockwise* and *counterclockwise* in naming (*R*) and (*S*) does not mean that light follows our naming rules.

If the (*R*) enantiomer rotates the plane 30° clockwise, the (*S*) enantiomer will rotate it 30° counterclockwise. If the (*R*) enantiomer rotates the plane 5° counterclockwise, the (*S*) enantiomer will rotate it 5° clockwise.

> We cannot predict which direction a particular enantiomer [either (*R*) or (*S*)] will rotate the plane of polarized light.

(*R*) and (*S*) are simply names, but the direction and magnitude of rotation are physical properties that must be measured.

5-4C Polarimetry

A **polarimeter** measures the rotation of polarized light. It has a tubular cell filled with a solution of the optically active material and a system for passing polarized light through the solution and measuring the rotation as the light emerges (Figure 5-14). The light from a sodium lamp is filtered so that it consists of just one wavelength (one color), because most compounds rotate different wavelengths of light by different amounts. The wavelength of light most commonly used for polarimetry is a yellow emission line in the spectrum of sodium, called the *sodium D line*.

Monochromatic (one-color) light from the source passes through a polarizing filter, then through the sample cell containing a solution of the optically active compound. Upon leaving the sample cell, the polarized light encounters another polarizing filter. This filter is movable, with a scale allowing the operator to read the angle between the axis of the second (analyzing) filter and the axis of the first (polarizing) filter. The operator rotates the analyzing filter until the maximum amount of light is transmitted, then reads the observed rotation from the protractor. The observed rotation is symbolized by α, the Greek letter alpha.

Compounds that rotate the plane of polarized light toward the right (clockwise as we look through the detector) are called **dextrorotatory,** from the Greek word

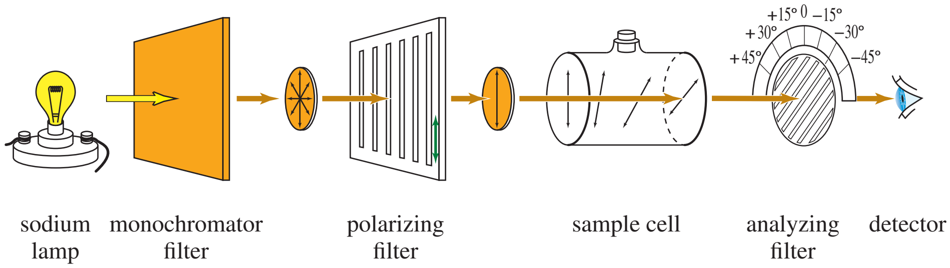


FIGURE 5-14 Schematic diagram of a polarimeter. The light originates at a source (usually a sodium lamp) and passes through a polarizing filter and the sample cell. An optically active solution rotates the plane of polarized light. The analyzing filter is another polarizing filter equipped with a protractor. It is turned until a maximum amount of light is observed, and the rotation is read from the protractor.

dexios, meaning "toward the right." Compounds that rotate the plane toward the left (counterclockwise) are called **levorotatory,** from the Latin word *laevus*, meaning "toward the left." These terms are sometimes abbreviated by a lowercase *d* or *l*. Using IUPAC notation, the direction of rotation is specified by the (+) or (−) sign of the rotation:

> Dextrorotatory (clockwise) rotations are (+) or (*d*).
> Levorotatory (counterclockwise) rotations are (−) or (*l*).

For example, the isomer of butan-2-ol that rotates the plane of polarized light clockwise is named (+)-butan-2-ol or *d*-butan-2-ol. Its enantiomer, (−)-butan-2-ol or *l*-butan-2-ol, rotates the plane counterclockwise by exactly the same amount.

You can see the principle of polarimetry by using two pairs of polarized sunglasses, a beaker, and some corn syrup or sugar solution. Wear one pair of sunglasses, look down at a light, and hold another pair of sunglasses above the light. Notice that the most light is transmitted through the two pairs of sunglasses when their axes are parallel. Very little light is transmitted when their axes are perpendicular.

Put syrup in the beaker, and hold the beaker above the bottom pair of sunglasses so that the light passes through one pair of sunglasses (the polarizing filter), then the beaker (the optically active sample), and then the other pair of sunglasses (the analyzing filter); see Figure 5-15. Again, check the angles giving maximum and minimum light transmission. Is the syrup solution dextrorotatory or levorotatory? Did you notice the color variation as you rotated the filter? You can see why just one color of light should be used for accurate work.

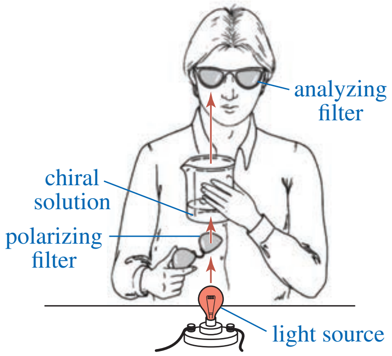


FIGURE 5-15 Apparatus for using a lightbulb and two pairs of polarized sunglasses as a simple polarimeter.

5-4D Specific Rotation

The angular rotation of polarized light by a chiral compound is a characteristic physical property of that compound, just like the boiling point or the density. The rotation (α) observed in a polarimeter depends on the concentration of the sample solution and the length of the cell, as well as the optical activity of the compound. For example, twice as concentrated a solution would give twice the original rotation. Similarly, a 20-cm cell gives twice the rotation observed using a similar concentration in a 10-cm cell.

To use the rotation of polarized light as a characteristic property of a compound, we must standardize the conditions for measurement. We define a compound's **specific rotation** $[\alpha]$ as the rotation found using a 10-cm (1-dm) sample cell and a concentration of 1 g/mL. Other cell lengths and concentrations may be used, as long as the observed rotation is divided by the path length of the cell (l) and the concentration (c).

$$[\alpha] = \frac{\alpha(\text{observed})}{c \cdot l}$$

where

$$\begin{aligned} \alpha(\text{observed}) &= \text{rotation observed in the polarimeter} \\ c &= \text{concentration in grams per mL} \\ l &= \text{length of sample cell (path length) in decimeters (dm)} \end{aligned}$$

A rotation depends on the wavelength of light used and on the temperature, so these data are given together with the rotation. We often see specific rotations given as $[\alpha]_{\mathrm{D}}^{25}$. The "25" means that the measurement was made at 25 °C, and the "D" means that the light used was the D line of the sodium spectrum.

SOLVED PROBLEM 5-4

When one of the enantiomers of butan-2-ol is placed in a polarimeter, the observed rotation is 4.05° counterclockwise. The solution was made by diluting 6.00 g of butan-2-ol to a total of 40.0 mL, and the solution was placed into a 200-mm polarimeter tube for the measurement. Determine the specific rotation for this enantiomer of butan-2-ol.

SOLUTION

Because it is levorotatory, this must be (−)-butan-2-ol. The concentration is 6.00 g per 40.0 mL = 0.150 g/mL, and the path length is 200 mm = 2.00 dm. The specific rotation is −13.5°.

$$[\alpha]_{\mathrm{D}}^{25} = \frac{-4.05°}{(0.150)(2.00)} = -13.5°$$

Without even measuring it, we can predict that the specific rotation of the other enantiomer of butan-2-ol will be

$$[\alpha]_{\mathrm{D}}^{25} = +13.5°$$

where the (+) refers to the clockwise direction of the rotation. This enantiomer would be called (+)-butan-2-ol. We could refer to this pair of enantiomers as (+)-butan-2-ol and (−)-butan-2-ol or as (*R*)-butan-2-ol and (*S*)-butan-2-ol.

Does this mean that (*R*)-butan-2-ol is the dextrorotatory isomer because it is named (*R*) and that (*S*)-butan-2-ol is levorotatory because it is named (*S*)? Not at all! The rotation of a compound, (+) or (−), is something that we measure in the polarimeter, depending on how the molecule interacts with light. The (*R*) and (*S*) nomenclature is our own artificial way of describing how the atoms are arranged in space.

In the laboratory, we can measure a rotation and see whether a particular substance is (+) or (−). On paper, we can determine whether a particular drawing is named (*R*) or (*S*). But it is difficult to predict whether a structure we call (*R*) will rotate polarized light clockwise or counterclockwise. Similarly, it is difficult to predict whether a dextrorotatory substance in a flask has the (*R*) or (*S*) configuration.

PROBLEM 5-8

A solution of 2.0 g of (−)-glyceraldehyde, $HOCH_2CHOHCHO$, in 5.0 mL of water was placed in a 100-mm cell. Using the sodium D line, a rotation of −1.74° was found at 25 °C. Determine the specific rotation of (−)-glyceraldehyde.

PROBLEM 5-9

A solution of 0.50 g of (+)-epinephrine (see Figure 5-16) dissolved in 10.0 mL of dilute aqueous HCl was placed in a 10-cm polarimeter tube. Using the sodium D line, the rotation was found to be +5.1° at 25 °C. Determine the specific rotation of epinephrine.

PROBLEM 5-10

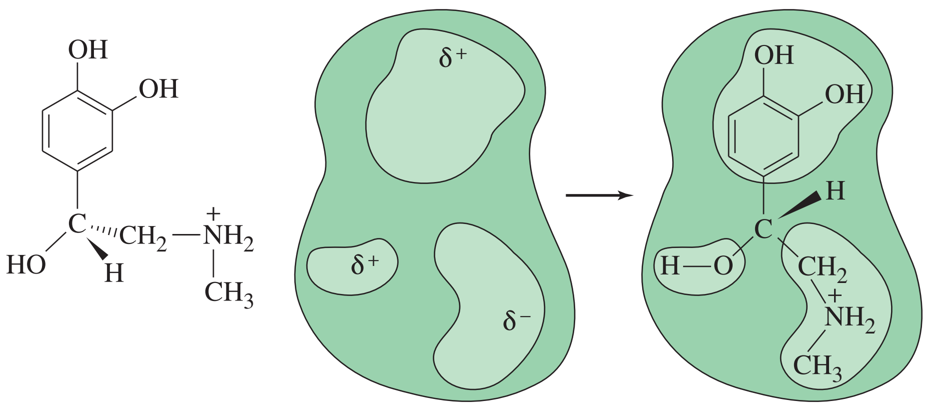
A chiral sample gives a rotation that is close to 180°. How can one tell whether this rotation is +180° or −180°?

5-5 Biological Discrimination of Enantiomers

If the direction of rotation of polarized light were the only difference between enantiomers, you might ask if the difference is important. However, biological systems commonly distinguish between enantiomers, and two enantiomers may have totally different biological properties. In fact, any **chiral probe** can distinguish between enantiomers, and a polarimeter is only one example of a chiral probe. Another example is your hand. If you needed to sort a box of gloves into right-handed gloves and left-handed gloves, you could distinguish between them by checking to see which ones fit your right hand.

Enzymes in living systems are chiral, and they are capable of distinguishing between enantiomers. Usually, only one enantiomer of a pair fits properly into the chiral active site of an enzyme. For example, the levorotatory form of epinephrine is one of the principal hormones secreted by the adrenal medulla. When synthetic epinephrine is given to a patient, the (−) form has the same stimulating effect as the natural hormone. The (+) form lacks this effect and is mildly toxic. Figure 5-16 shows a simplified picture of how only the (−) enantiomer fits into the enzyme's active site.

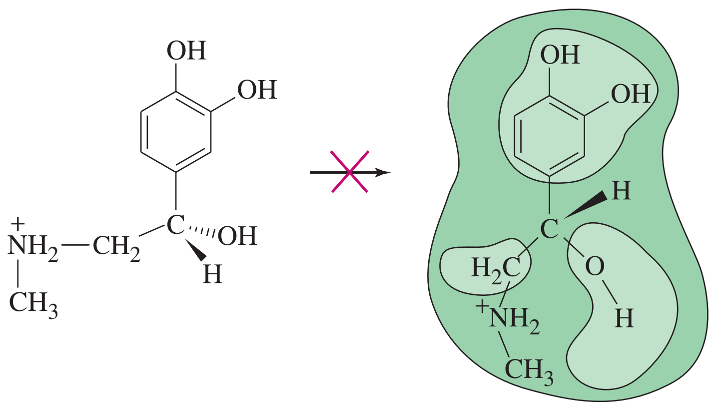
Biological systems are capable of distinguishing between the enantiomers of many different chiral compounds. In general, just one of the enantiomers produces the characteristic effect; the other either produces no effect or has a different effect. Even your nose is capable of distinguishing between some enantiomers. For example,



(*R*)-(−)-epinephrine
natural epinephrine

enzyme active site

enzyme–substrate complex



(*S*)-(+)-epinephrine
unnatural epinephrine

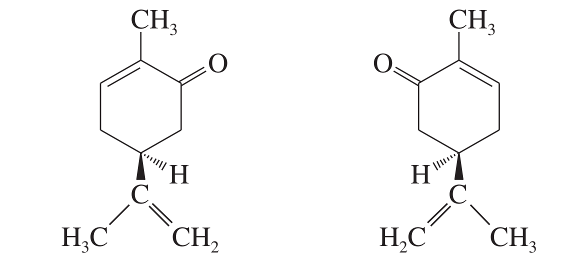
does not fit the enzyme's active site

FIGURE 5-16 Chiral recognition of epinephrine by an enzyme. Only the levorotatory enantiomer fits into the active site of the enzyme.

Application: Biochemistry

Enzymes can exist as two enantiomers, although only one enantiomer is found in nature. In 1992, Stephen Kent and coworkers reported the synthesis of both enantiomers of an enzyme that cuts peptide substrates, and showed for the first time that each enzyme acts only on the corresponding enantiomeric peptide substrate.

(−)-carvone is the fragrance associated with spearmint oil; (+)-carvone has the tangy odor of caraway seed. The receptor sites for the sense of smell must be chiral, therefore, just as the active sites in most enzymes are chiral. In general, enantiomers do not interact identically with other chiral molecules, whether or not they are of biological origin.



(+)-carvone (caraway seed) (−)-carvone (spearmint)

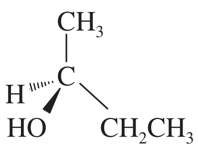
PROBLEM 5-11

If you had the two enantiomers of carvone in unmarked bottles, could you use just your nose and a polarimeter to determine

(a) whether it is the (+) or (−) enantiomer that smells like spearmint?
(b) whether it is the (*R*) or (*S*) enantiomer that smells like spearmint?
(c) With the information given in the drawings of carvone above, what can you add to your answers to (a) and (b)?

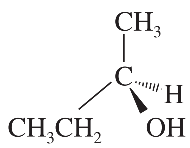
5-6 Racemic Mixtures

Suppose we had a mixture of equal amounts of (+)-butan-2-ol and (−)-butan-2-ol. The (+) isomer would rotate polarized light clockwise with a specific rotation of +13.5°, and the (−) isomer would rotate the polarized light counterclockwise by exactly the same amount. We would observe a rotation of zero, just as though butan-2-ol were achiral. A solution of equal amounts of two enantiomers, so that the mixture is optically inactive, is called a **racemic mixture.** Sometimes a racemic mixture is called a **racemate,** a **(±) pair,** or a **(*d,l*) pair.** A racemic mixture is symbolized by placing (±) or (*d,l*) in front of the name of the compound. For example, racemic butan-2-ol would be symbolized by "(±)-butan-2-ol" or "(*d,l*)-butan-2-ol."



(*S*)-(+)-butan-2-ol
+13.5° rotation

and

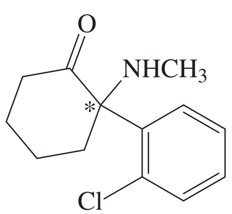


(*R*)-(−)-butan-2-ol
−13.5° rotation

A racemic mixture contains equal amounts of the two enantiomers.

Application: Drugs

Many drugs currently on the market are racemic mixtures. Ketamine, for example, is a potent anesthetic agent, but its use is limited because it is hallucinogenic (making it a drug of abuse widely known as "K"). The (*S*) isomer is responsible for the anesthetic effects, and the (*R*) isomer causes the hallucinogenic effects.

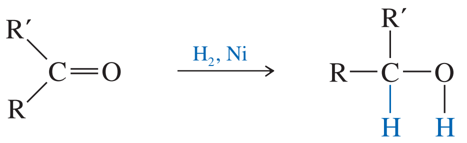


ketamine

You might think that a racemic mixture would be unusual because it requires exactly equal amounts of the two enantiomers. This is not the case, however. Many reactions lead to racemic products, especially when an achiral molecule is converted to a chiral molecule.

> A reaction that uses optically inactive reactants and catalysts cannot produce a product that is optically active. Any chiral product must be formed as a racemic mixture.

For example, hydrogen adds across the C═O double bond of a ketone to produce an alcohol.



Because the carbonyl group is flat, a simple ketone such as butan-2-one is achiral. Hydrogenation of butan-2-one gives butan-2-ol, a chiral molecule (Figure 5-17). This reaction involves adding hydrogen atoms to the C═O carbon atom and oxygen atom. If the hydrogen atoms are added to one face of the double bond, the (*S*) enantiomer results. Addition of hydrogen to the other face forms the (*R*) enantiomer. It is equally probable for hydrogen to add to either face of the double bond, and equal amounts of the (*R*) and (*S*) enantiomers are formed.

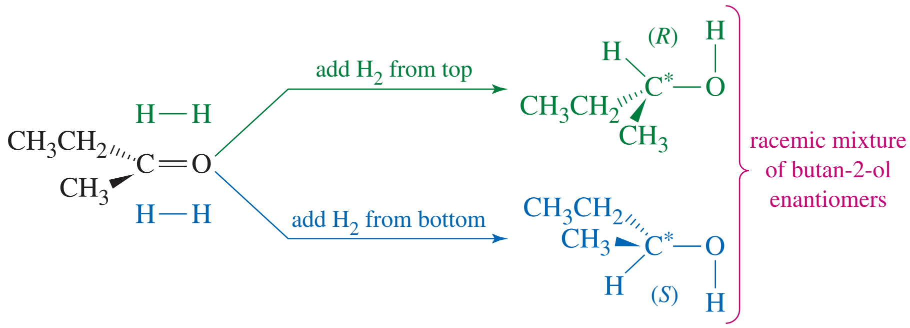


FIGURE 5-17 Hydrogenation of butan-2-one forms racemic butan-2-ol. Hydrogen adds to either face of the double bond. Addition of H_2 to one face gives the (*R*) product, while addition to the other face gives the (*S*) product.

Logically, it makes sense that optically inactive reagents and catalysts cannot form optically active products. If the starting materials and reagents are optically inactive, there is no reason for the dextrorotatory product to be favored over a levorotatory one or vice versa. The (+) product and the (−) product are favored equally, and they are formed in equal amounts: a racemic mixture.

5-7 Enantiomeric Excess and Optical Purity

Sometimes we deal with mixtures that are neither optically pure (all one enantiomer) nor racemic (equal amounts of two enantiomers). In these cases, we specify the **optical purity (o.p.)** of the mixture. The optical purity of a mixture is defined as the ratio of its rotation to the rotation of a pure enantiomer. For example, if we have some [mostly (+)] butan-2-ol with a specific rotation of +9.72°, we compare this rotation with the +13.5° rotation of the pure (+) enantiomer.

$$\text{o.p.} = \frac{\text{observed rotation}}{\text{rotation of pure enantiomer}} \times 100\% = \frac{9.72°}{13.5°} \times 100\% = 72.0\%$$

The **enantiomeric excess (e.e.)** is a more common method for expressing the relative amounts of enantiomers in a mixture. To compute the enantiomeric excess of a mixture, we calculate the *excess* of the predominant enantiomer as a percentage of the entire mixture. For a chemically pure compound, the calculation of enantiomeric excess generally gives the same result as the calculation of optical purity, and we often use the two terms interchangeably. Algebraically, we use the following formula:

$$\text{o.p.} = \text{e.e.} = \frac{|d - l|}{d + l} \times 100\% = \frac{\text{(excess of one over the other)}}{\text{(entire mixture)}} \times 100\%$$

The units cancel out in the calculation of either e.e. or o.p., so these formulas can be used whether the amounts of the enantiomers are expressed in concentrations, grams, or percentages. For the butan-2-ol mixture just described, the optical purity of 72% (+) implies that $d - l = 72\%$, and we know that $d + l = 100\%$. Adding the equations gives $2d = 172\%$. We conclude that the mixture contains 86% of the d or (+) enantiomer and 14% of the l or (−) enantiomer.

Application: Drugs

Several racemic drugs have become available as pure active enantiomers. For example, the drug Nexium® (for controlling acid reflux) contains just the active enantiomer of the racemic mixture in Prilosec®.

SOLVED PROBLEM 5-5

Calculate the e.e. and the specific rotation of a mixture containing 6.0 g of (+)-butan-2-ol and 4.0 g of (−)-butan-2-ol.

SOLUTION

In this mixture, there is a 2.0 g excess of the (+) isomer and a total of 10.0 g, for an e.e. of 20%. We can envision this mixture as 80% racemic [4.0 g(+) and 4.0 g(−)] and 20% pure (+).

$$\text{o.p.} = \text{e.e.} = \frac{|6.0 - 4.0|}{6.0 + 4.0} = \frac{2.0}{10.0} = 20\%$$

The specific rotation of enantiomerically pure (+)-butan-2-ol is +13.5°. The rotation of this mixture is

$$\begin{aligned}\text{observed rotation} &= (\text{rotation of pure enantiomer}) \times (\text{o.p.}) \\ &= (+13.5°) \times (20\%) = +2.7°\end{aligned}$$

PROBLEM 5-12

When optically pure (R)-2-bromobutane is heated with water, butan-2-ol is the product. The reaction forms twice as much (S)-butan-2-ol as (R)-butan-2-ol. Calculate the e.e. and the specific rotation expected for the product.

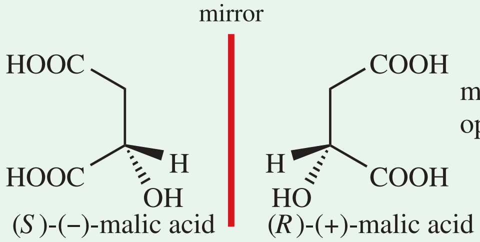
PROBLEM 5-13

A chemist finds that the addition of (+)-epinephrine to the catalytic reduction of butan-2-one (Figure 5-17) gives a product that is slightly optically active, with a specific rotation of +0.45°. Calculate the percentages of (+)-butan-2-ol and (−)-butan-2-ol formed in this reaction.

FOCUS Enantiomers and Racemic Mixtures

Enantiomers used to be called **optical isomers**, defined as compounds identical in all physical properties except that they rotate plane-polarized light by equal amounts but in opposite directions. For example, the two enantiomers of malic acid have these properties.

melting point = 102 °C
optical rotation = –28°



(S)-(−)-malic acid (R)-(+)-malic acid

melting point = 102 °C
optical rotation = +28°

Today, we now define enantiomers in terms of structure: enantiomers are different compounds that are nonsuperimposable mirror images of each other. As drawn above, the enantiomers of malic acid are reflected in the imaginary mirror between them. If these were molecular models, picking up one and flipping it over to try to superimpose the atoms would put one OH up and the other OH down. No amount of twisting or rotating would achieve overlap of all of the atoms. They are different structures.

Enantiomers can be mixed in any proportion, but the 50:50 mixture is an important special case, called a **racemic mixture**.

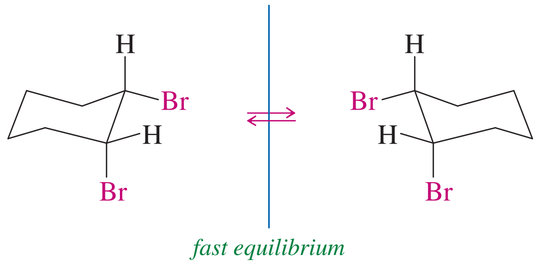
A 50:50 mixture of (*S*)- and (*R*)-malic acid, denoted as (*R*,*S*)-(±)-malic acid, has these properties: $\begin{cases}\text{melting point} = 131\ °\text{C} \\ \text{optical rotation} = 0°\end{cases}$

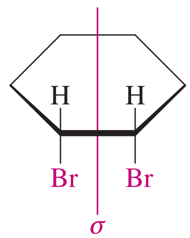
A racemic mixture can and often does have physical properties different from each individual pure enantiomer. For malic acid, the individual enantiomers have a melting point of 102 °C, but the racemic mixture has a melting point of 131 °C. However, a racemic mixture must always have 0° optical rotation, because the optical rotations of the individual enantiomers cancel each other.

5-8 Chirality of Conformationally Mobile Systems

Let's consider whether *cis*-1,2-dibromocyclohexane is chiral. If we did not know about chair conformations, we might draw a flat cyclohexane ring. With a flat ring, the molecule has an internal mirror plane of symmetry (σ), and it is achiral.

But we know the ring is puckered into a chair conformation with one bromine atom axial and one equatorial. A chair conformation of *cis*-1,2-dibromocyclohexane and its mirror image are shown below. These two mirror-image structures are nonsuperimposable. You may be able to see the difference more easily if you make models of these two chiral conformations.





Does this mean that *cis*-1,2-dibromocyclohexane is chiral? No, it does not, because the chair–chair interconversion is rapid at room temperature. If we had a bottle of just the conformation on the left, the molecules would quickly undergo chair–chair interconversions. Because the two mirror-image conformations interconvert and have identical energies, any sample of *cis*-1,2-dibromocyclohexane must contain equal amounts of the two mirror images. Similarly, nearly all achiral compounds can exist in transient chiral conformations that are in equilibrium with their mirror-image conformations. For a compound to be considered chiral, it must be possible, at least in theory, to isolate a sample containing just one enantiomer.

We consider a molecule to be achiral if its chiral conformations are in rapid equilibrium with their mirror-image conformations, such that we cannot purify a sample of just one enantiomer.

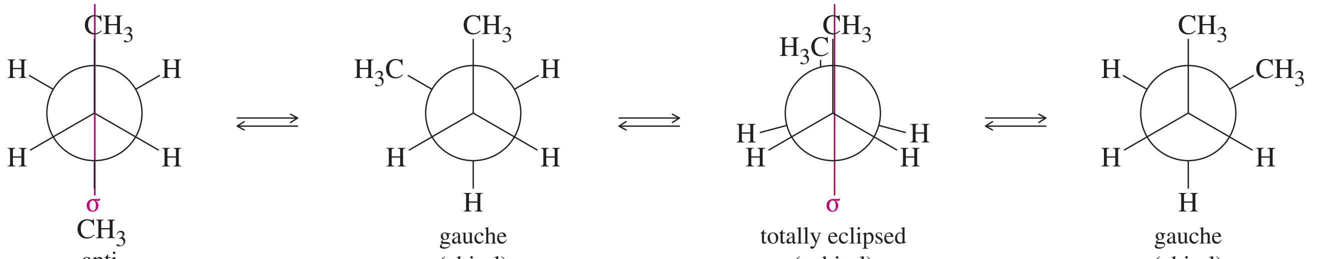
cis-1,2-Dibromocyclohexane appears to exist as a racemic mixture, but with a major difference: It is impossible to isolate an optically active sample of *cis*-1,2- dibromocyclohexane. We could have predicted the correct result by imagining that the cyclohexane ring is flat.

This finding leads to a general principle we can use with conformationally mobile systems:

> To determine whether a conformationally mobile molecule is chiral, consider its most symmetric conformation.

We can consider cyclohexane rings as though they were flat (the most symmetric conformation), and we should consider straight-chain compounds in their most symmetric conformations (often an eclipsed conformation).

Organic compounds commonly exist as rapidly interconverting chiral conformations. Even ethane is chiral in its skew conformations. When we speak of chirality, however, we intend to focus on observable, persistent properties rather than transient conformations. For example, butane exists in gauche conformations that are chiral, but they quickly interconvert. They are in equilibrium with the anti and totally eclipsed conformations, which are both symmetric, implying that butane must be achiral.



In contrast, we can draw any of the conformations of (*R*)-2-bromobutane (Figure 5-4), and they are all chiral. None of them can interconvert with their mirror images because the (*R*) asymmetric carbon cannot interconvert with its (*S*) mirror image.

PROBLEM-SOLVING HINT

Nearly all organic compounds have chiral conformations.

A chiral compound has no symmetric (achiral) conformations, and none of its conformations can interconvert with their mirror images.

An achiral compound can (and usually does) rotate through chiral conformations, but the chiral conformations quickly interconvert with their mirror images.

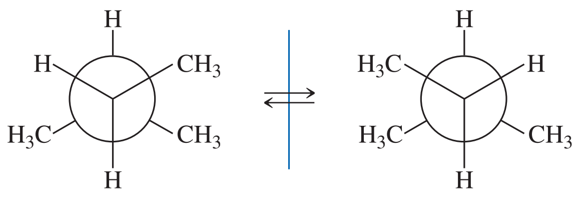
SOLVED PROBLEM 5-6

Draw each compound in its most stable conformation(s). Then draw it in its most symmetric conformation, and determine whether it is chiral.

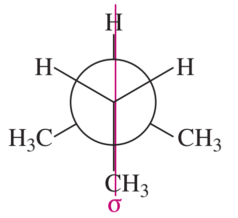
(a) 2-methylbutane

SOLUTION

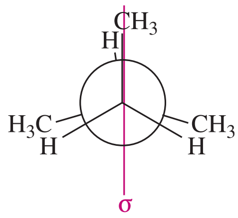
The most stable conformations of 2-methylbutane are two mirror-image conformations. These conformations are nonsuperimposable, but they can interconvert by rotation around the central bond. So they are not enantiomers.



2-Methylbutane has two symmetric conformations: Either of these conformations is sufficient to show that 2-methylbutane is achiral.



symmetric conformation

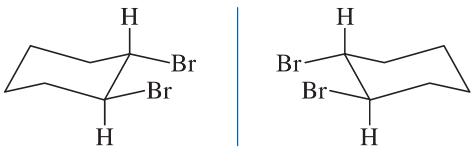


symmetric conformation

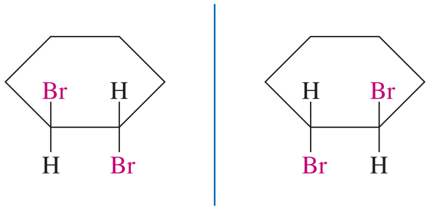
(b) *trans*-1,2-dibromocyclohexane

SOLUTION

We can draw two nonsuperimposable mirror images of the most stable chair conformation of *trans*-1,2-dibromocyclohexane with both bromines equatorial. These structures cannot interconvert by ring-flips or other rotations about bonds, however. They are mirror-image isomers: enantiomers.



This molecule's chirality is more apparent when drawn in its most symmetric conformation. Drawn flat, the two mirror-image structures of *trans*-1,2-dibromocyclohexane are still nonsuperimposable. This compound is inherently chiral, and no conformational changes can interconvert the two enantiomers.



PROBLEM-SOLVING HINT

Consider the most symmetric accessible conformation. You can also consider the most stable conformation and see if it can interconvert with its mirror image.

PROBLEM 5-14

1. Make a model of each compound, draw it in its most symmetric conformation, and determine whether it is capable of showing optical activity.

- **(a)** 1-bromo-1-chloroethane
- **(b)** 1-bromo-2-chloroethane
- **(c)** 1,2-dichloropropane
- **(d)** *cis*-1,3-dibromocyclohexane
- **(e)** *trans*-1,3-dibromocyclohexane
- **(f)** *trans*-1,4-dibromocyclohexane

2. Star (*) each asymmetric carbon atom in part (1), label each as (*R*) or (*S*), and compare your result from part (1) with the prediction you would make based on the asymmetric carbons.

5-9 Chiral Compounds Without Asymmetric Atoms

Most chiral organic compounds have at least one asymmetric carbon atom. Some compounds are chiral because they have another asymmetric atom, such as phosphorus, sulfur, or nitrogen, serving as a chiral center. Some compounds are chiral even though they have no asymmetric atoms at all. In these types of compounds, special characteristics of the molecules' shapes lead to chirality.

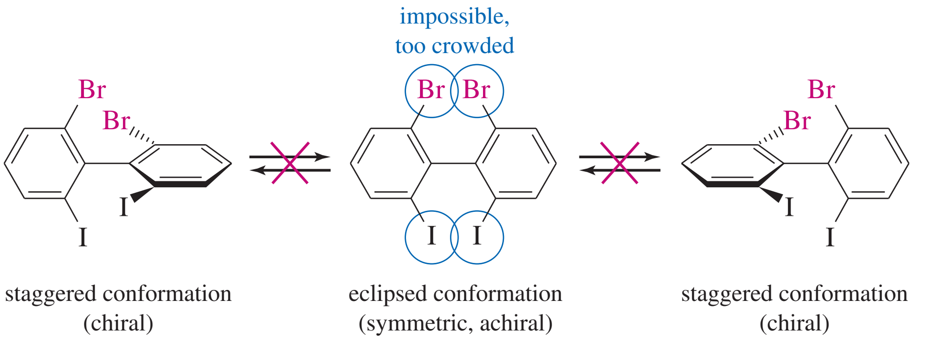


FIGURE 5-18 Three conformations of a biphenyl. This biphenyl cannot pass through its symmetric conformation because there is too much crowding of the iodine and bromine atoms. The molecule is "locked" into one of the two chiral, enantiomeric, staggered conformations.

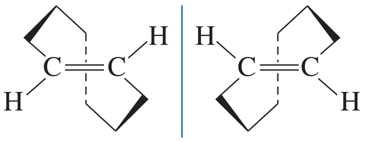


FIGURE 5-19 Conformational enantiomerism. *trans*-Cyclooctene is strained, unable to achieve a symmetric planar conformation. It is locked into one of these two enantiomeric conformations. Either pure enantiomer is optically active, with $[\alpha] = \pm 430°$.

5-9A Conformational Enantiomerism

Some molecules are so bulky or so highly strained that they cannot easily convert from one chiral conformation to the mirror-image conformation. They cannot achieve the most symmetric conformation because it has too much steric strain or ring strain. Because these molecules are "locked" into a conformation, we must evaluate the individual locked-in conformation to determine whether the molecule is chiral.

Figure 5-18 shows three conformations of a sterically crowded derivative of biphenyl. The center drawing shows the molecule in its most symmetric conformation. This conformation is planar, and it has a mirror plane of symmetry. If the molecule could achieve this conformation, or even pass through it for an instant, it would not be optically active. This planar conformation is very high in energy, however, because the iodine and bromine atoms are too large to be forced so close together. The molecule is *conformationally locked*. It can exist only in one of the two staggered conformations shown on the left and right. These conformations are nonsuperimposable mirror images, and they do not interconvert. They are enantiomers, and they can be separated and isolated. Each of them is optically active, and they have equal and opposite specific rotations.

Even a simple strained molecule can show conformational enantiomerism (Figure 5-19). *trans*-Cyclooctene is the smallest stable *trans*-cycloalkene, and it is strained. If *trans*-cyclooctene existed as a planar ring, even for an instant, it could not be chiral. Make a molecular model of *trans*-cyclooctene, however, and you will see that it cannot exist as a planar ring. Its ring is folded into the three-dimensional structure pictured in Figure 5-19. The mirror image of this structure is different, and *trans*-cyclooctene is a chiral molecule. In fact, the enantiomers of *trans*-cyclooctene have been separated and characterized, and they are optically active.

5-9B Allenes

Allenes are compounds that contain the C═C═C unit, with two C═C double bonds meeting at a single carbon atom. The parent compound, propadiene, has the common name *allene*.

$H_2C{=}C{=}CH_2$

allene

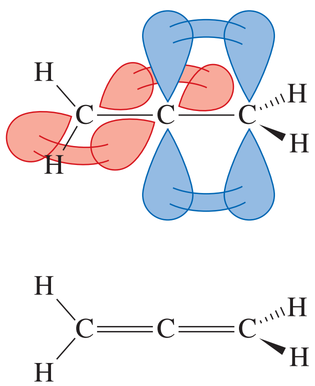


FIGURE 5-20 Structure of allene. The two ends of the allene molecule are perpendicular.

In allene, the central carbon atom is *sp* hybridized and linear (Section 1-15), and the two outer carbon atoms are sp^2 hybridized and trigonal. We might imagine that the whole molecule lies in a plane, but this is not correct. The central *sp* hybrid carbon atom must use different *p* orbitals to form the pi bonds with the two outer carbon atoms. The two unhybridized *p* orbitals on the *sp* hybrid carbon atom are perpendicular, so the two pi bonds must also be perpendicular. Figure 5-20 shows the bonding and three-dimensional structure of allene. Allene itself is achiral. If you make a model of its mirror image, you will find it identical with the original molecule. If we add some substituents to allene, however, the molecule may be chiral.

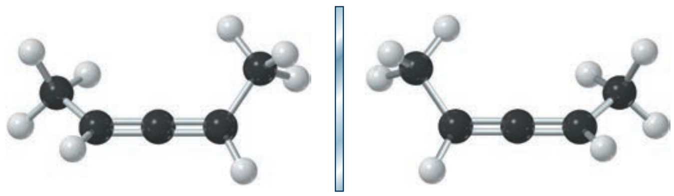
Make a model of the following compound:

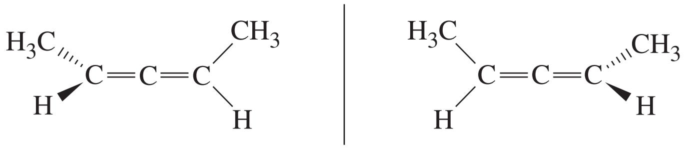
$$\overset{1}{C}H_3-\overset{2}{C}H=\overset{3}{C}=\overset{4}{C}H-\overset{5}{C}H_3$$

penta-2,3-diene

Carbon atom 3 is the *sp* hybrid allene-type carbon atom. Carbons 2 and 4 are both sp^2 and planar, but their planes are perpendicular to each other. None of the carbon atoms is attached to four different atoms, so there is no asymmetric carbon atom. Nevertheless, penta-2,3-diene is chiral, as you should see from your models and from the following drawings of the enantiomers.

mirror





enantiomers of penta-2,3-diene

PROBLEM-SOLVING HINT

Dienes are compounds with two double bonds. In the name, each double bond is given the lower number of its two carbon atoms. *Allenes* are dienes with the two double bonds next to each other, joined at one carbon atom. An allene is chiral if each end has two distinct substituents.

PROBLEM 5-15

Draw three-dimensional representations of the following compounds. Which have asymmetric carbon atoms? Which have no asymmetric carbons but are chiral anyway? Use your models for parts (a) through (d) and any others that seem unclear.

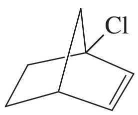
(a) $ClHC=C=CHCl$
1,3-dichloropropadiene

(b) $ClHC=C=CHCH_3$
1-chlorobuta-1,2-diene

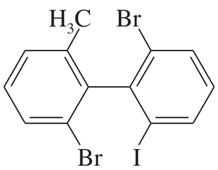
(c) $ClHC=C=C(CH_3)_2$
1-chloro-3-methylbuta-1,2-diene

(d) $ClHC=CH-CH=CH_2$
1-chlorobuta-1,3-diene

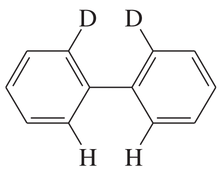
(e)



(f)



(g)



PROBLEM-SOLVING HINT

All asymmetric carbons are stereocenters, but not all stereocenters are asymmetric carbons.

5-10 Fischer Projections

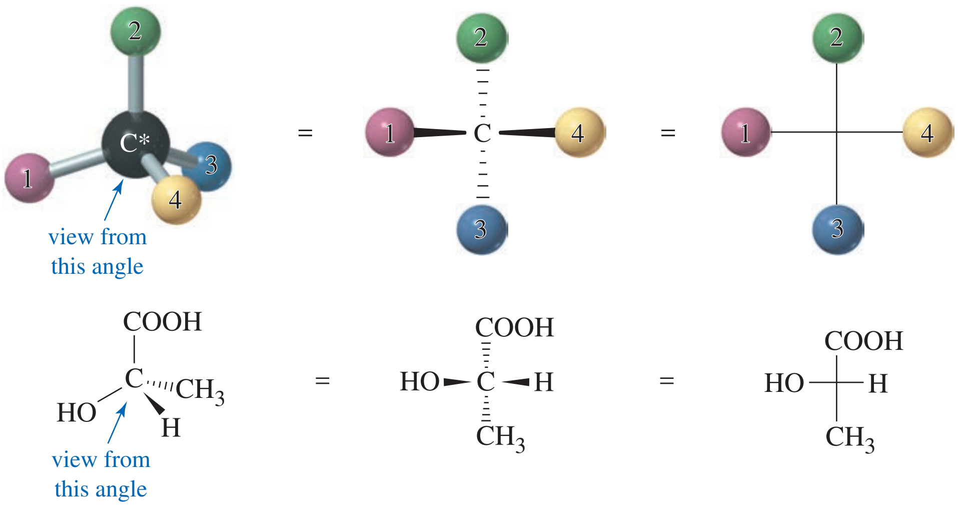
We have been using dashed lines and wedges to indicate perspective in drawing the stereochemistry of asymmetric carbon atoms. When we draw molecules with several asymmetric carbons, perspective drawings become time-consuming and cumbersome. In addition, the complicated drawings make it difficult to see the similarities and differences in groups of stereoisomers.

At the turn of the 20th century, Emil Fischer was studying the stereochemistry of sugars (Chapter 23), which contain as many as seven asymmetric carbon atoms. To draw these structures in perspective would have been difficult, and to pick out minor stereochemical differences in the drawings would have been nearly impossible. Fischer developed a symbolic way of drawing asymmetric carbon atoms, allowing them to be drawn rapidly. The **Fischer projection** also facilitates comparison of stereoisomers, holding them in their most symmetric conformation and emphasizing any differences in stereochemistry. We learn to use Fischer projections because they clarify stereochemical differences and allow fast and accurate comparison of structures.

5-10A Drawing Fischer Projections

The Fischer projection looks like a cross, with the asymmetric carbon (usually not drawn in) at the point where the lines cross. The horizontal lines are taken to be wedges—that is, bonds that project out toward the viewer. The vertical lines are taken to project away from the viewer, as dashed lines. Figure 5-21 shows the perspective implied by the Fischer projection. The center drawing, with the wedged horizontal bonds looking like a bow tie, illustrates why this projection is sometimes called the "bow-tie convention." Problem 5-16 should help you to visualize how the Fischer projection is used.

FIGURE 5-21 Perspective in a Fischer projection. The Fischer projection uses a cross to represent an asymmetric carbon atom. The horizontal lines project toward the viewer, and the vertical lines project away from the viewer.



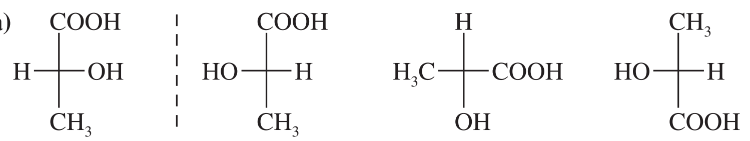
(*S*)-lactic acid perspective drawing

(*S*)-lactic acid Fischer projection

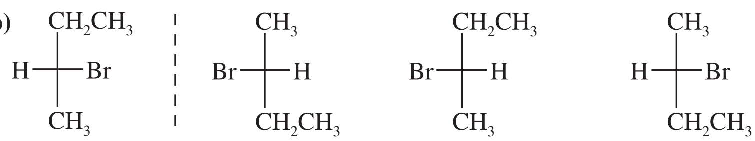
PROBLEM 5-16

For each set of examples, make a model of the first structure, and indicate the relationship of each of the other structures to the first structure. Examples of relationships: same compound, enantiomer, structural isomer.

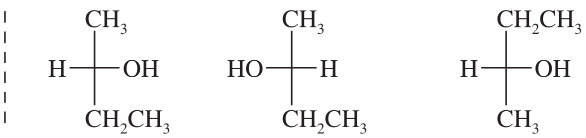
(a)



(b)

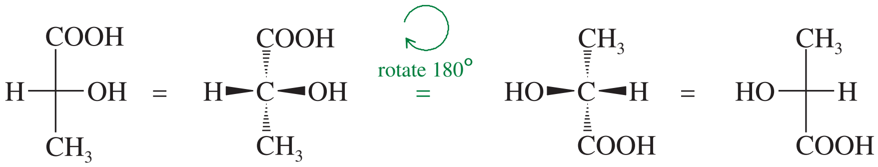


(c) (*R*)-butan-2-ol



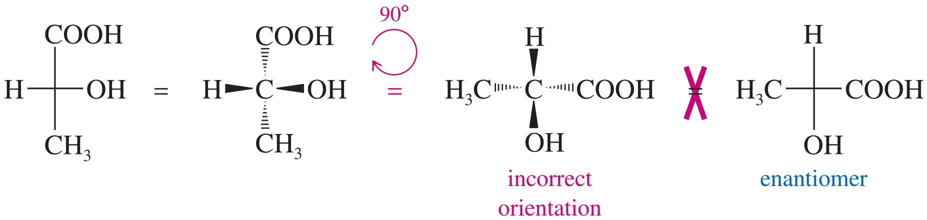
In working Problem 5-16, you may have noticed that Fischer projections that differ by a 180° rotation are the same. When we rotate a Fischer projection by 180°, the vertical (dashed line) bonds still end up vertical, and the horizontal (wedged) lines still end up horizontal. The "horizontal lines forward, vertical lines back" convention is maintained.

Rotation by 180° is allowed.



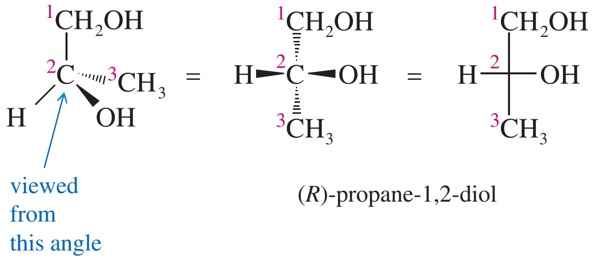
On the other hand, if we were to rotate a Fischer projection by 90°, we would change the configuration and confuse the viewer. The original projection has the vertical groups back (dashed lines) and the horizontal groups forward. When we rotate the projection by 90°, the vertical bonds become horizontal and the horizontal bonds become vertical. The viewer assumes that the horizontal bonds come forward and that the vertical bonds go back. The viewer sees a different molecule (actually, the enantiomer of the original molecule).

A 90° rotation is NOT allowed.



In comparing Fischer projections, we cannot rotate them by 90°, and we cannot flip them over. Either of these operations gives an incorrect representation of the molecule. The Fischer projection must be kept in the plane of the paper, and it may be rotated only by 180°.

The final rule for drawing Fischer projections helps to ensure that we do not rotate the drawing by 90°. This rule is that the carbon chain is drawn along the vertical line of the Fischer projection, usually with the IUPAC numbering from top to bottom. In most cases, this numbering places the most highly oxidized carbon substituent at the top. For example, to represent (*R*)-propane-1,2-diol with a Fischer projection, we should arrange the three carbon atoms along the vertical. C1 is placed at the top, and C3 at the bottom.



(*R*)-propane-1,2-diol

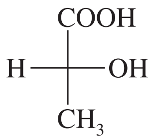
PROBLEM 5-17

Draw a Fischer projection for each compound. Remember that the cross represents an asymmetric carbon atom, and the carbon chain should be along the vertical, with the IUPAC numbering from top to bottom.

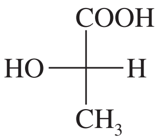
(a) (*S*)-propane-1,2-diol
(b) (*R*)-2-bromobutan-1-ol
(c) (*S*)-1,2-dibromobutane
(d) (*R*)-butan-2-ol
(e) (*R*)-glyceraldehyde, $HO{-}{}^{3}CH_2{-}{}^{2}CH(OH){-}{}^{1}CHO$

PROBLEM-SOLVING HINT

Interchanging any two groups on a Fischer projection (or on a perspective drawing) inverts the configuration of that asymmetric carbon from (*R*) to (*S*) or from (*S*) to (*R*).



(*R*)-lactic acid



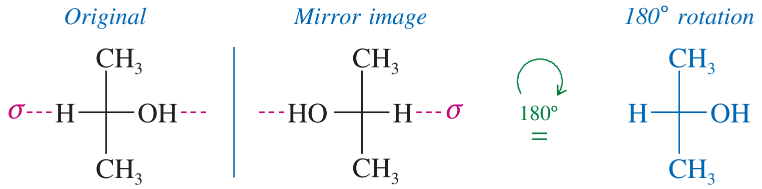
(*S*)-lactic acid

5-10B Drawing Mirror Images of Fischer Projections

How should we draw the mirror image of a molecule drawn in Fischer projection? With our perspective drawings, the rule was to reverse left and right while keeping the other directions (up and down, front and back) in their same positions. This rule still applies to Fischer projections. Interchanging the groups on the horizontal part of the cross reverses left and right while leaving the other directions unchanged.

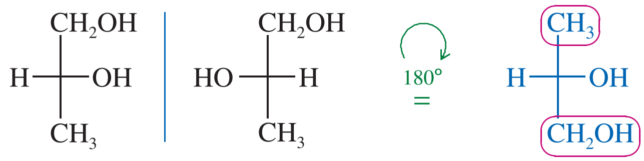
Testing for enantiomerism is particularly simple using Fischer projections. If the Fischer projections are properly drawn (carbon chain along the vertical), and if the mirror image cannot be made to look the same as the original structure with a 180° rotation in the plane of the paper, the two mirror images are enantiomers. In the following examples, any groups that fail to superimpose after a 180° rotation are circled in red.

propan-2-ol



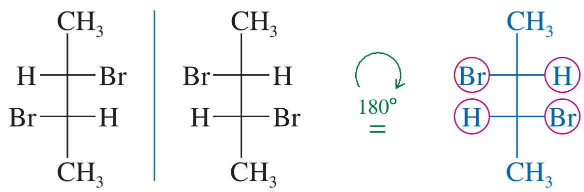
These mirror images are the same. Propan-2-ol is achiral.

(*R*)-propane-1,2-diol



These mirror images are different. Propane-1,2-diol is chiral.

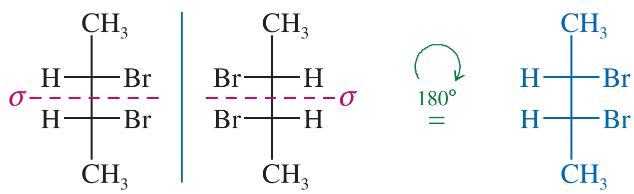
(2*S*,3*S*)-2,3-dibromobutane



These mirror images are different. This structure is chiral.

Mirror planes of symmetry are particularly easy to identify from the Fischer projection because this projection is normally the most symmetric conformation. In the first preceding example (propan-2-ol) and in the following example [(2*S*,3*R*)-2,3-dibromobutane], the symmetry planes are indicated in red; these molecules with symmetry planes cannot be chiral.

(2*S*,3*R*)-2,3-dibromobutane

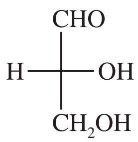


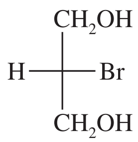
These mirror images are the same. This structure is achiral.

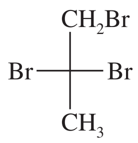
PROBLEM 5-18

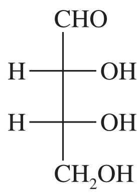
For each Fischer projection,

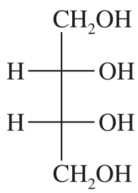
1. make a model.
2. draw the mirror image.
3. determine whether the mirror image is the same as, or different from, the original structure.
4. draw any mirror planes of symmetry that are apparent from the Fischer projections.

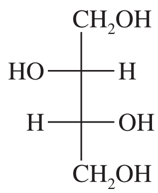
(a)


(b)


(c)


(d)


(e)


(f)


PROBLEM-SOLVING HINT

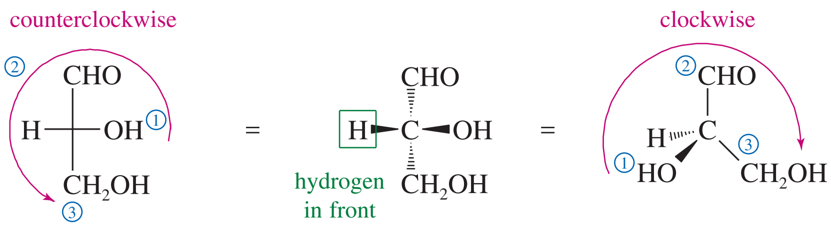
Using Fischer projections, we can easily solve difficult stereochemical problems that would be intractable using perspective drawings. Most problems involving two or more asymmetric carbons are more easily solved using Fischer projections.

5-10C Assigning (*R*) and (*S*) Configurations from Fischer Projections

The Cahn–Ingold–Prelog convention (Section 5-3) can be applied to structures drawn using Fischer projections. Let's review the two rules for assigning (*R*) and (*S*): (1) Assign priorities to the groups bonded to the asymmetric carbon atom; (2) put the lowest-priority group (usually H) in back, and draw an arrow from group 1 to group 2 to group 3. Clockwise is (*R*), and counterclockwise is (*S*).

The (*R*) or (*S*) configuration can also be determined directly from the Fischer projection, without having to convert it to a perspective drawing. The lowest-priority atom is usually hydrogen. In the Fischer projection, the carbon chain is along the vertical line, so the hydrogen atom is usually on the horizontal line and projects out in front. Once we have assigned priorities, we can draw an arrow from group 1 to group 2 to group 3 and see which way it goes. If the molecule were turned around so that the hydrogen would be in back [as in the definition of (*R*) and (*S*)], the arrow would rotate in the other direction. By mentally turning the arrow around (or simply applying the rule backward), we can assign the configuration.

As an example, consider the Fischer projection formula of one of the enantiomers of glyceraldehyde. First priority goes to the —OH group, followed by the —CHO group and the —CH_2OH group. The hydrogen atom receives the lowest priority. The arrow from group 1 to group 2 to group 3 appears counterclockwise in the Fischer projection. If the molecule is turned over so that the hydrogen is in back, the arrow becomes clockwise, making this the (*R*) enantiomer of glyceraldehyde.



Fischer projection (*R*)-(+)-glyceraldehyde

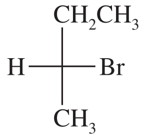
perspective drawing (*R*)-(+)-glyceraldehyde

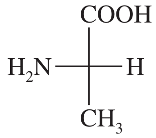
PROBLEM-SOLVING HINT

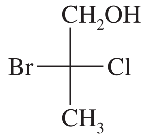
When naming (*R*) and (*S*) from Fischer projections with the hydrogen on a horizontal bond (toward you instead of away from you), just apply the normal rules backward.

PROBLEM 5-19

For each Fischer projection, label each asymmetric carbon atom as (*R*) or (*S*).
(a)–(f) the structures in Problem 5-18

(g)


(h)


(i)

(careful—no hydrogen)

SUMMARY Fischer Projections and Their Use

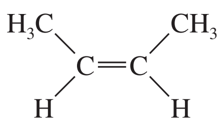
1. They are most useful for compounds with two or more asymmetric carbon atoms.
2. Asymmetric carbons are at the centers of crosses.
3. The vertical lines project away from the viewer; the horizontal lines project toward the viewer (like a bow tie ).
4. The carbon chain is placed along the vertical, with the IUPAC numbering from top to bottom. In most cases, this places the more oxidized end (the carbon with the most bonds to O or halogen) at the top.
5. The entire projection can be rotated 180° (but not 90°) in the plane of the paper without changing its stereochemistry.
6. Interchanging any two groups on an asymmetric carbon (for example, those on the horizontal line) inverts its stereochemistry.

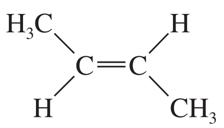
5-11 Diastereomers

We have defined *stereoisomers* as isomers whose atoms are bonded together in the same order but differ in how the atoms are directed in space. We have also considered enantiomers (mirror-image isomers) in detail. All other stereoisomers are classified as **diastereomers,** which are defined as *stereoisomers that are not mirror images*. Most diastereomers are either geometric isomers or compounds containing two or more chiral centers.

5-11A Cis-trans Isomerism on Double Bonds

We have already seen one class of diastereomers: the **cis-trans isomers,** or **geometric isomers.** For example, there are two isomers of but-2-ene:

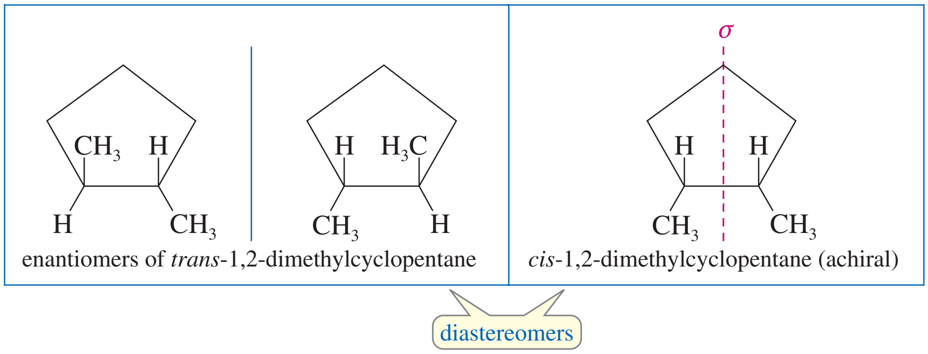

cis-but-2-ene


trans-but-2-ene

These stereoisomers are not mirror images of each other, so they are not enantiomers. They are diastereomers.

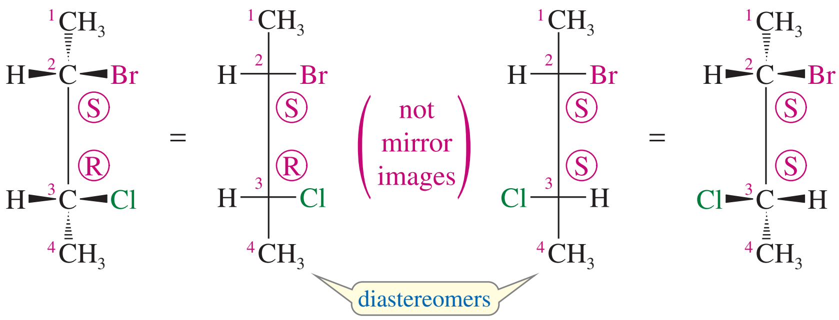
5-11B Cis-trans Isomerism on Rings

Cis-trans isomerism is also possible when a ring is present. *Cis*- and *trans*-1,2-dimethylcyclopentane are geometric isomers, and they are diastereomers. The trans diastereomer has an enantiomer, but because the cis diastereomer has an internal mirror plane of symmetry, it must be achiral.



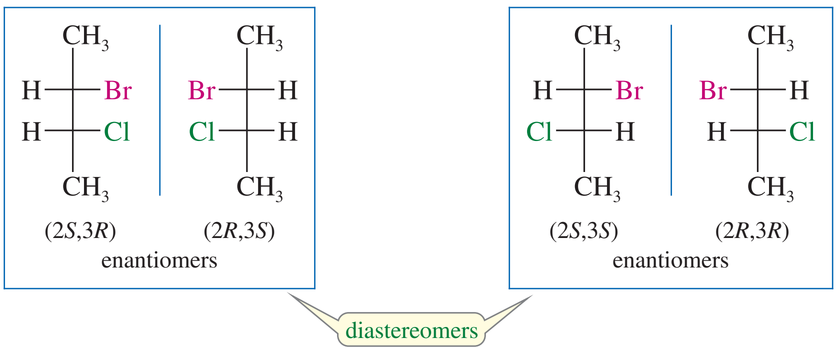
5-11C Diastereomers of Molecules with Two or More Chiral Centers

Apart from geometric isomers, most other compounds that show diastereomerism have two or more chiral centers, usually asymmetric carbon atoms. For example, 2-bromo-3-chlorobutane has two asymmetric carbon atoms, and it exists in two diastereomeric forms (shown next). Make molecular models of these two stereoisomers.



These two structures are not the same; they are stereoisomers because they differ in the orientation of their atoms in space. They are not enantiomers, however, because they are not mirror images of each other: C2 has the (*S*) configuration in both structures, while C3 is (*R*) in the structure on the left and (*S*) in the structure on the right. The C3 carbon atoms are mirror images of each other, but the C2 carbon atoms are not. If these two compounds were mirror images of each other, both asymmetric carbons would have to be mirror images of each other.

Because these compounds are stereoisomers but not enantiomers, they must be diastereomers. In fact, both of these diastereomers are chiral and each has an enantiomer. Thus, there is a total of four stereoisomeric 2-bromo-3-chlorobutanes: two pairs of enantiomers. Either member of one pair of enantiomers is a diastereomer of either member of the other pair.

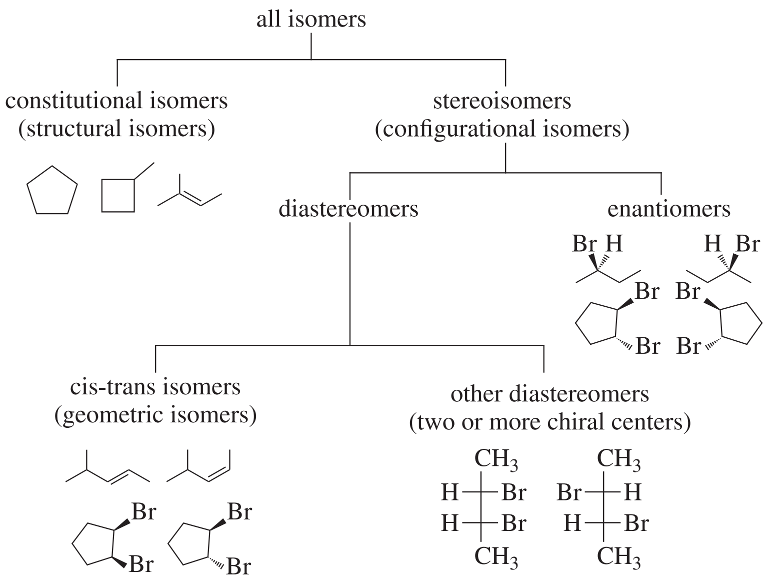


PROBLEM-SOLVING HINT

When a compound contains two or more chiral centers (usually asymmetric carbons), consider drawing it as a Fischer projection. The stereochemical differences we see in the structures at left are not as clear in perspective drawings, which are also much harder to draw.

We have now seen all the types of isomers we need to study, and we can diagram their relationships and summarize their definitions.

SUMMARY Types of Isomers



Isomers are different compounds with the same molecular formula.

Constitutional isomers are isomers that differ in the order in which atoms are bonded together. Constitutional isomers are sometimes called **structural isomers** because they have different connections among their atoms.

Stereoisomers are isomers that differ only in the orientation of the atoms in space.

Enantiomers are mirror-image isomers.

Diastereomers are stereoisomers that are not mirror images of each other.

Cis-trans isomers (geometric isomers) are diastereomers that differ in their cis-trans arrangement on a ring or double bond.

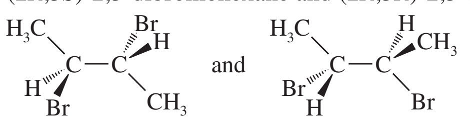
PROBLEM 5-20

For each pair, give the relationship between the two compounds. Making models will be helpful.

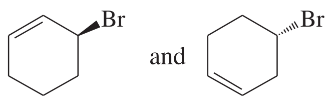
(a) (2*R*,3*S*)-2,3-dibromohexane and (2*S*,3*R*)-2,3-dibromohexane

(b) (2*R*,3*S*)-2,3-dibromohexane and (2*R*,3*R*)-2,3-dibromohexane

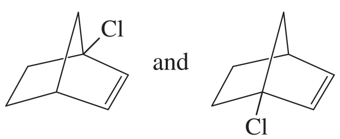
(c)



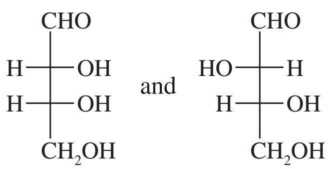
(d)



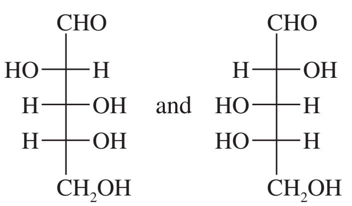
(e)



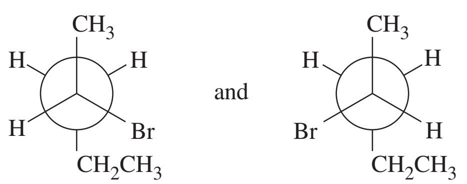
(f)



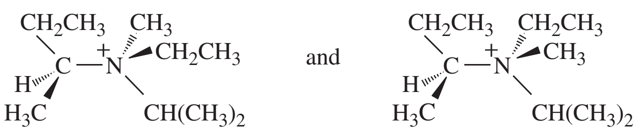
(g)



(h)

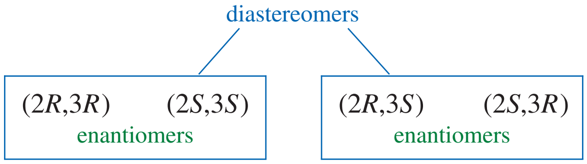


(i)



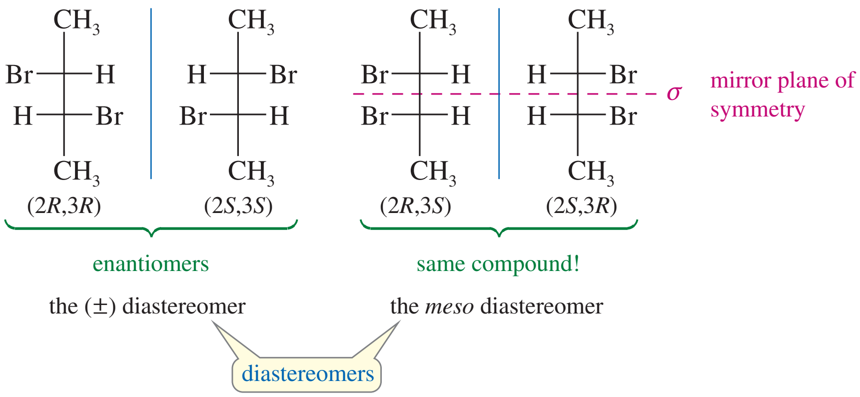
5-12 Stereochemistry of Molecules with Two or More Asymmetric Carbons

In the preceding section, we saw that there are four stereoisomers (two pairs of enantiomers) of 2-bromo-3-chlorobutane. These four isomers are simply all the permutations of (*R*) and (*S*) configurations at the two asymmetric carbon atoms, C2 and C3:



A compound with *n* asymmetric carbon atoms might have as many as has 2^n stereoisomers. This formula is called the **2^n rule,** where *n* is the number of chiral centers (usually asymmetric carbon atoms). The 2^n rule suggests that we should look for a *maximum* of 2^n stereoisomers. We may not always find 2^n isomers, especially when two of the asymmetric carbon atoms have identical substituents.

2,3-Dibromobutane has fewer than 2^n stereoisomers. It has two asymmetric carbons (C2 and C3), so the 2^n rule predicts a maximum of four stereoisomers. The four permutations of (*R*) and (*S*) configurations at C2 and C3 are shown next. Make molecular models of these structures to compare them.



There are only three stereoisomers of 2,3-dibromobutane because two of the four structures are identical. The diastereomer on the right is achiral, having a mirror plane of symmetry. The asymmetric carbon atoms have identical substituents, and the one with (*R*) configuration reflects into the other having (*S*) configuration. It seems almost as though the molecule were a racemic mixture within itself.

5-13 Meso Compounds

Compounds that are achiral even though they have asymmetric carbon atoms are called **meso compounds.** The (2*R*,3*S*) isomer of 2,3-dibromobutane is a meso compound; most meso compounds have this kind of symmetric structure, with two similar halves of the molecule having opposite configurations. In speaking of the two diastereomers of 2,3-dibromobutane, the symmetric one is called the *meso diastereomer*, and the chiral one is called the (±) *diastereomer*, because one enantiomer is (+) and the other is (−).

> MESO COMPOUND: An achiral compound that has chiral centers (usually asymmetric carbons).

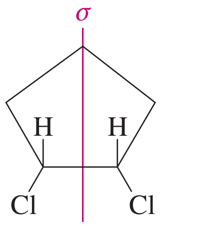
The term *meso* (Greek, "middle") was used to describe an achiral member of a set of diastereomers, some of which are chiral. The optically inactive isomer seemed to be in the

PROBLEM-SOLVING HINT

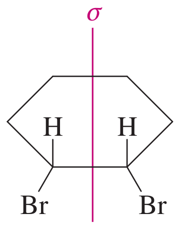
A meso compound with two chiral centers will be (*R*,*S*) or (*S*,*R*) because the chiral centers must be mirror images of each other, reflected across the internal mirror plane.

"middle" between the dextrorotatory and levorotatory isomers. The definition just given ("an achiral compound with chiral centers") is nearly as complete, and more easily applied, especially when you remember that chiral centers are usually asymmetric carbon atoms.

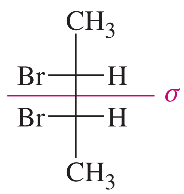
We have already seen other meso compounds, although we have not yet called them that. For example, the cis isomer of 1,2-dichlorocyclopentane has two asymmetric carbon atoms, yet it is achiral. Thus, it is a meso compound. *cis*-1,2-Dibromocyclohexane is not symmetric in its chair conformation, but it consists of equal amounts of two enantiomeric chair conformations in a rapid equilibrium. We are justified in looking at the molecule in its symmetric flat conformation to show that it is achiral and meso. For acyclic compounds, the Fischer projection helps to show the symmetry of meso compounds.



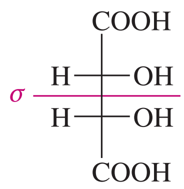
cis-1,2-dichlorocyclopentane



cis-1,2-dibromocyclohexane



meso-2,3-dibromobutane

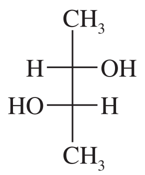


meso-tartaric acid

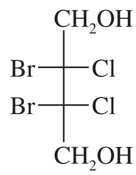
SOLVED PROBLEM 5-7

Determine which of the following compounds are chiral. Star (*) any asymmetric carbon atoms, and draw in any mirror planes. Label any meso compounds. (Use your molecular models to follow along.)

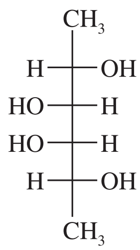
(a)



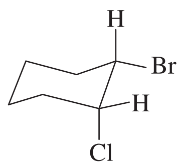
(b)



(c)

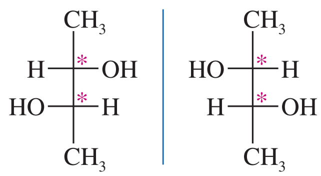


(d)



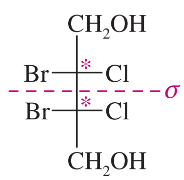
SOLUTION

(a)



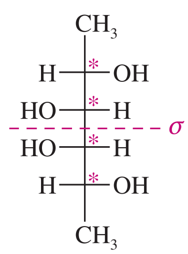
This compound does *not* have a plane of symmetry, and we suspect that it is chiral. Drawing the mirror image shows that it is nonsuperimposable on the original structure. These are the enantiomers of a chiral compound.

(b)



meso

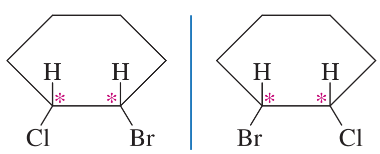
(c)



meso

Both (b) and (c) have mirror planes of symmetry and are achiral. Because they have asymmetric carbon atoms yet are achiral, they are meso.

(d)



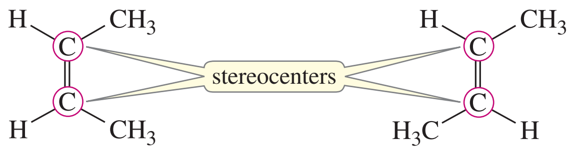
Drawing this compound in its most symmetric conformation (flat) shows that it does not have a mirror plane of symmetry. When we draw the mirror image, we find it to be an enantiomer.

SOLVED PROBLEM 5-8

One source defines a meso compound as "an achiral compound with stereocenters." Why is this a poor definition?

SOLUTION

A stereocenter is an atom at which the interchange of two groups gives a stereoisomer. Stereocenters include both chiral centers and double-bonded carbons giving rise to cis-trans isomers. For example, the isomers of but-2-ene are achiral and they contain stereocenters (circled), so they meet this definition. They have no chiral diastereomers, however, so they are not correctly called meso.

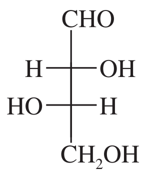


PROBLEM 5-21

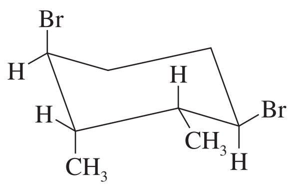
Which of the following compounds are chiral? Draw each compound in its most symmetric conformation, star (*) any asymmetric carbon atoms, and draw any mirror planes. Label any meso compounds. You may use Fischer projections if you prefer.

(a) *meso*-2,3-dibromo-2,3-dichlorobutane
(b) (±)-2,3-dibromo-2,3-dichlorobutane
(c) (2*R*,3*S*)-2-bromo-3-chlorobutane
(d) (2*R*,3*S*)-2,3-dibromobutane
(e) (*R*,*R*)-2,3-dibromobutane

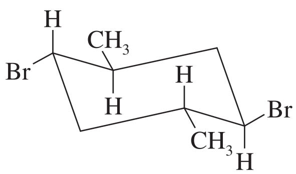
(f)



(g)



(*h)



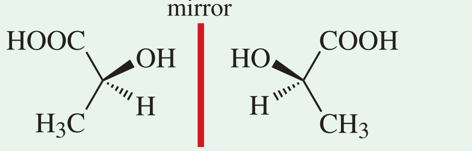
PROBLEM-SOLVING HINT

To draw all the stereoisomers of a given structure, first identify all the stereocenters. Then systematically go through all the permutations that result when the stereocenters are in their two configurations. For example, if there are three chiral centers, you write RRR, RRS, RSR, SRR, RSS, SRS, SSR, and SSS. In a Fischer projection, you can start with all the substituents on one side and go through the same process moving them to the other side. If the molecule has symmetry that creates meso structures, you must eliminate the duplicates.

FOCUS Multiple Chiral Centers

A compound with one chiral center has a nonsuperimposable mirror image, an enantiomer, with identical physical properties except for giving the opposite rotation of plane-polarized light. The two enantiomers can be mentally interconverted by inverting the configuration at the chiral center.

mirror

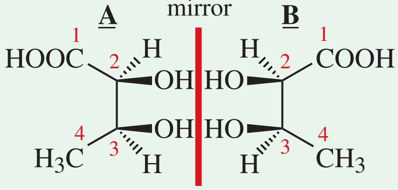


(*S*)-lactic acid
melting point = 54 °C
optical rotation = +2.7°

(*R*)-lactic acid
melting point = 54 °C
optical rotation = –2.7°

The compound 2,3-dihydroxybutanoic acid has two chiral centers. Compound **A** has a mirror image **B** that is nonsuperimposable; compounds **A** and **B** are enantiomers. Similarly, compounds **C** and **D** are enantiomers, but they are not mirror images of **A** and **B**. Compounds **C** and **D** are stereoisomers of **A** and **B** that are not enantiomers; they are **diastereomers** of **A** and **B**.

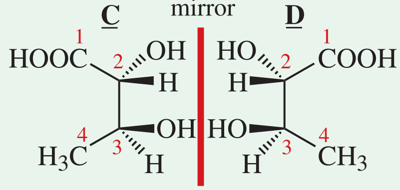
A mirror **B**



(2*S*,3*S*) melting point (mp) = 64 °C, optical rotation (or) = +11°
(2*R*,3*R*) mp = 64 °C, or = –11°

***A* and *B* are enantiomers.**

C mirror **D**



(2*R*,3*S*) mp = 75 °C, or = +19°
(2*S*,3*R*) mp = 75 °C, or = –19°

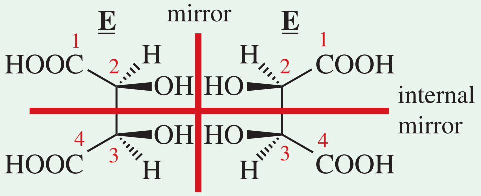
***C* and *D* are enantiomers.**

Enantiomers have the same physical properties except for the opposite direction of rotation. Diastereomers can and usually do have different properties. The maximum number of stereoisomers = 2^n where n = number of chiral centers.

When drawing structures of compounds with multiple chiral centers:

- *inverting all chiral centers makes the enantiomer;*
- *inverting at least one chiral center while keeping at least one chiral center the same makes a diastereomer.*

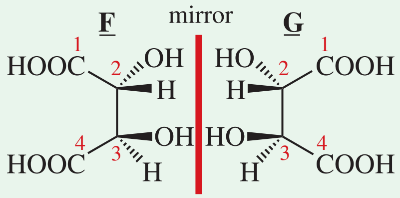
In compounds with multiple chiral centers, a special case can occur with compounds that have an internal mirror plane of symmetry, e.g. tartaric acid. A **meso compound** has two or more chiral centers but is identical with its mirror image. It is not optically active. *The presence of a meso compound reduces the maximum number of stereoisomers.* Tartaric acid has a total of only three stereoisomers, not four as in the case of 2,3-dihydroxybutanoic acid above.



(2*R*,3*S*) mp = 166 °C or = 0°

(2*S*,3*R*) mp = 166 °C or = –0°

These are THE SAME COMPOUND, a MESO compound. Name: meso-tartaric acid.



(2*R*,3*R*) mp = 172 °C or = +12°

(2*S*,3*S*) mp = 172 °C or = –12°

F *and* **G** *are enantiomers.*

PROBLEM-SOLVING HINT

In part (e), the carbon bearing the OH group is not asymmetric, but it can be a stereocenter if the methyl groups are cis to each other.

PROBLEM 5-22

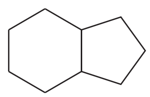
Draw all the distinct stereoisomers for each structure. Show the relationships (enantiomers, diastereomers, etc.) between the isomers. Label any meso isomers, and draw any mirror planes of symmetry.

(a) CH_3—CHCl—CHOH—COOH

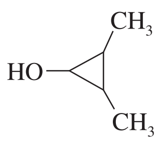
(b) tartaric acid, HOOC—CHOH—CHOH—COOH

(c) HOOC—CHBr—CHOH—CHOH—COOH

(d)



(e)



5-14 Absolute and Relative Configuration

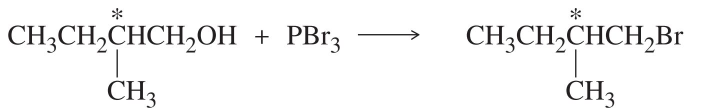
Throughout our study of stereochemistry, we have drawn three-dimensional representations, and we have spoken of asymmetric carbons having the (*R*) or (*S*) configuration. These ways of describing the configuration of a chiral center are *absolute*; that is, they give the actual orientation of the atoms in space. We say that these methods specify the **absolute configuration** of the molecule. For example, given the name "(*R*)-butan-2-ol," any chemist can construct an accurate molecular model or draw a three-dimensional representation.

ABSOLUTE CONFIGURATION: The detailed stereochemical picture of a molecule, including how the atoms are arranged in space. Alternatively, the (*R*) or (*S*) configuration at each chiral center.

Chemists have determined the absolute configurations of many chiral compounds since 1951, when X-ray crystallography was first used to find the orientation of atoms in space. Before 1951, there was no way to link the stereochemical drawings with the actual enantiomers and their observed rotations. No absolute configurations were known. It was possible, however, to correlate the configuration of one compound with another and to show that two compounds had the same or opposite configurations. When we convert one compound into another using a reaction that does not break bonds at the asymmetric carbon atom, we know that the product must have the same **relative configuration** as the reactant, even if we cannot determine the absolute configuration of either compound.

RELATIVE CONFIGURATION: The experimentally determined relationship between the configurations of two molecules, even though we may not know the absolute configuration of either.

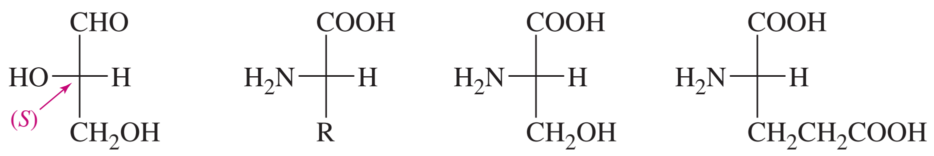
For example, optically active 2-methylbutan-1-ol reacts with PBr_3 to give optically active 1-bromo-2-methylbutane. None of the bonds to the asymmetric carbon atom are broken in this reaction, so the product must have the same configuration at the asymmetric carbon as the starting material does.



(+)-2-methylbutan-1-ol $[\alpha]_D^{25} = +5.8°$ (−)-1-bromo-2-methylbutane $[\alpha]_D^{25} = -4.0°$

We say that (+)-2-methylbutan-1-ol and (−)-1-bromo-2-methylbutane have the same relative configuration, even though we don't have the foggiest idea whether either of these is (*R*) or (*S*) unless we relate them to a compound whose absolute configuration has been established by X-ray crystallography.

Before the advent of X-ray crystallography, several systems were used to compare the relative configurations of chiral compounds with those of standard compounds. Only one of these systems is still in common use today: the **D–L system,** also known as the *Fischer–Rosanoff convention*. The configurations of sugars and amino acids were related to the enantiomers of glyceraldehyde. Compounds with the same relative configuration as (+)-glyceraldehyde were assigned the D prefix, and those with the relative configuration of (−)-glyceraldehyde were given the L prefix.



L-(−)-glyceraldehyde an L-amino acid L-(−)-serine L-(+)-glutamic acid

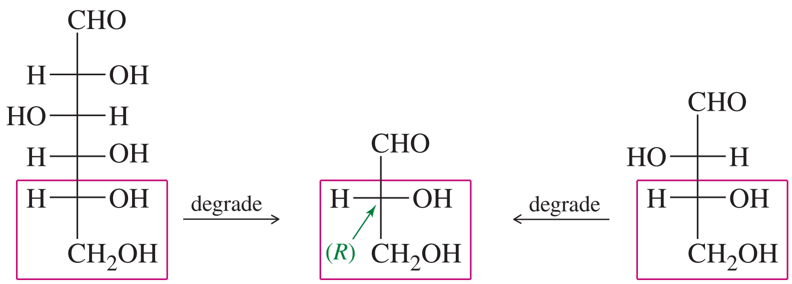
We now know the absolute configurations of the glyceraldehyde enantiomers: The (+) enantiomer has the (*R*) configuration, with the hydroxy (OH) group on the right in the Fischer projection. The (−) enantiomer has the (*S*) configuration, with the hydroxy

PROBLEM-SOLVING HINT

With chiral cyclic compounds, *cis* and *trans* prefixes give relative configurations. In the name *cis*-1-bromo-2-methylcyclohexane, *cis* indicates the relative positions of the methyl group and the bromine atom. It does not specify the absolute configuration of either chiral center, however, and we might correctly draw either cis enantiomer. The name (1*S*,2*R*)-1-bromo-2-methylcyclohexane specifies the absolute configuration of one of the cis enantiomers with only one possible structure.

group on the left. Most naturally occurring amino acids have the L configuration, with the amino (NH_2) group on the left in the Fischer projection.

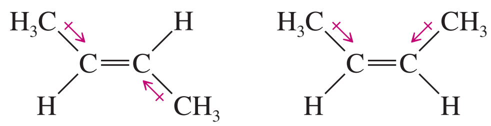
Sugars have several asymmetric carbons, but they can all be degraded to glyceraldehyde by oxidizing them from the aldehyde end. (We discuss these reactions in Chapter 23.) Most naturally occurring sugars degrade to (+)-glyceraldehyde, so they are given the D prefix. This means that the bottom asymmetric carbon of the sugar has its hydroxy (OH) group on the right in the Fischer projection.



D-(+)-glucose D-(+)-glyceraldehyde D-(−)-threose

5-15 Physical Properties of Diastereomers

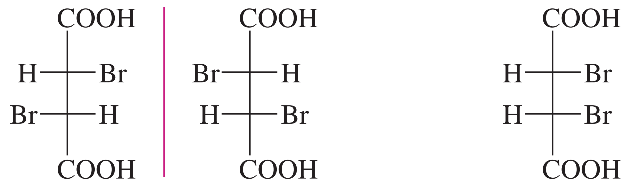
We have seen that enantiomers have identical physical properties except for the direction in which they rotate polarized light. Diastereomers, on the other hand, generally have different physical properties. For example, consider the diastereomers of but-2-ene (shown next). The symmetry of *trans*-but-2-ene causes the dipole moments of the bonds to cancel. The dipole moments in *cis*-but-2-ene do not cancel but add together to create a molecular dipole moment. The dipole–dipole attractions of *cis*-but-2-ene give it a higher boiling point than *trans*-but-2-ene.



$\mu = 0$
trans-but-2-ene
bond dipoles cancel
bp = 0.9 °C

$\mu = 0.33$ D
cis-but-2-ene
vector sum dipole
bp = 3.7 °C

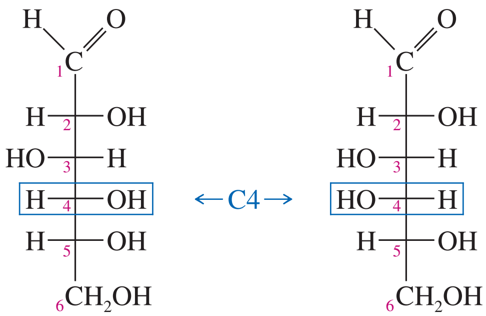
Diastereomers that are not geometric isomers also have different physical properties. The two diastereomers of 2,3-dibromosuccinic acid have melting points that differ by nearly 100 °C!



(+) and (−)-2,3-dibromosuccinic acid
mp for either is 158 °C

meso-2,3-dibromosuccinic acid
mp 256 °C

Most of the common sugars are diastereomers of glucose. All these diastereomers have different physical properties. For example, glucose and galactose are diastereomeric sugars that differ only in the stereochemistry of one asymmetric carbon atom, C4.



D-(+)-glucose, mp 148 °C D-(+)-galactose, mp 167 °C

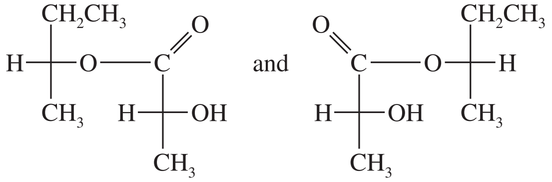
Because diastereomers have different physical properties, we can separate them by ordinary means such as distillation, recrystallization, and chromatography. As we will see in the next section, the separation of enantiomers is a more difficult process.

PROBLEM 5-23

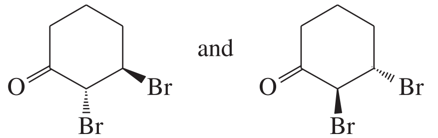
Which of the following pairs of compounds could be separated by recrystallization or distillation?

(a) *meso*-tartaric acid and (±)-tartaric acid (HOOC—CHOH—CHOH—COOH)

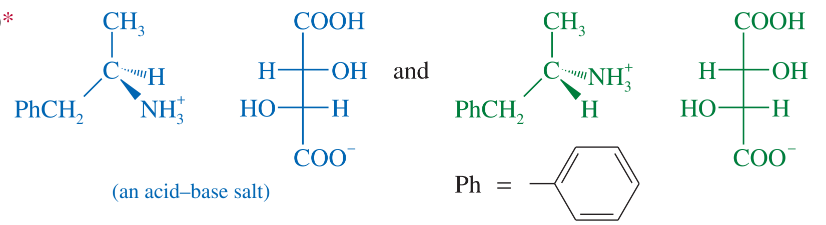
(b)



(c)



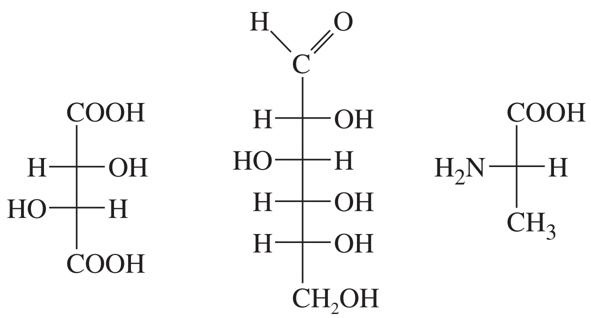
(d)*



(an acid–base salt)

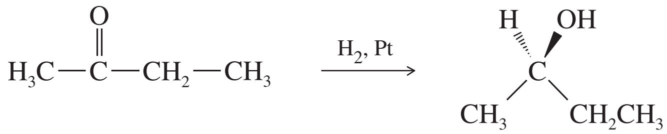
5-16 Resolution of Enantiomers

Pure enantiomers of optically active compounds are often obtained by isolation from biological sources. Most optically active molecules are found as only one enantiomer in living organisms. For example, pure (+)-tartaric acid can be isolated from the precipitate formed by yeast during the fermentation of wine. Pure (+)-glucose is obtained from many different sugar sources, such as grapes, sugar beets, sugarcane, and honey. Alanine is a common amino acid found in protein as the pure (+) enantiomer.



L-(+)-tartaric acid (2*R*,3*R*)-tartaric acid D-(+)-glucose L-(+)-alanine

When a chiral compound is synthesized from achiral reagents, however, a racemic mixture of enantiomers results. For example, we saw that the reduction of butan-2-one (achiral) to butan-2-ol (chiral) gives a racemic mixture:



butan-2-one → (*R*)-butan-2-ol + (*S*)-butan-2-ol

If we need one pure enantiomer of butan-2-ol, we must find a way of separating it from the other enantiomer. The separation of enantiomers is called **resolution,** and it is a different process from the usual physical separations. A chiral probe is necessary for the resolution of enantiomers; such a chiral compound or apparatus is called a **resolving agent.**

Wine sediments often contain salts of tartaric acid. This wine cork bears crystals of the potassium salt of L-(+)-tartaric acid.

In 1848, Louis Pasteur noticed that a salt of racemic (±)-tartaric acid crystallizes into mirror-image crystals. Using a microscope and a pair of tweezers, he physically separated the enantiomeric crystals. He found that solutions made from the "left-handed" crystals rotate polarized light in one direction and solutions made from the "right-handed" crystals rotate polarized light in the opposite direction. Pasteur had accomplished the first artificial resolution of enantiomers. Unfortunately, few racemic compounds crystallize as separate enantiomers, and other methods of separation are required.

5-16A Chemical Resolution of Enantiomers

The traditional method for resolving a racemic mixture into its enantiomers is to use an enantiomerically pure natural product that bonds with the compound to be resolved. When the enantiomers of the racemic compound bond to the pure resolving agent, a pair of diastereomers results. The diastereomers are separated, then the resolving agent is cleaved from the separated enantiomers.

Let's consider how we might resolve a racemic mixture of (*R*)- and (*S*)-butan-2-ol. We need a resolving agent that reacts with an alcohol and that is readily available in an enantiomerically pure state. A carboxylic acid combines with an alcohol to form an ester. Although we have not yet studied the chemistry of esters (Chapter 21), the following equation shows how an acid and an alcohol can combine with the loss of water to form an ester.

$$\underset{\text{acid}}{R{-}\overset{\overset{\displaystyle O}{\|}}{C}{-}OH} \;+\; \underset{\text{alcohol}}{R'{-}OH} \;\xrightleftharpoons{(H^+\ \text{catalyst})}\; \underset{\text{ester}}{R{-}\overset{\overset{\displaystyle O}{\|}}{C}{-}O{-}R'} \;+\; \underset{\text{water}}{H{-}O{-}H}$$

For our resolving agent, we need an optically active chiral acid to react with butan-2-ol. Any winery can provide large amounts of pure (+)-tartaric acid. Figure 5-22 shows that diastereomeric esters are formed when (*R*)- and (*S*)-butan-2-ol react with (+)-tartaric acid. We can represent the reaction schematically as follows:

$$\begin{array}{c}\boxed{(R)\text{- and }(S)\text{-butan-2-ol}}\\ \text{plus}\\ \boxed{(R,R)\text{-tartaric acid}}\end{array} \xrightarrow{H^+} \boxed{\begin{array}{c}(R)\text{-2-butyl}\\(R,R)\text{-tartrate}\end{array}} + \boxed{\begin{array}{c}(S)\text{-2-butyl}\\(R,R)\text{-tartrate}\end{array}}$$

diastereomers, *not* mirror images

An illustration of Louis Pasteur working in the laboratory. He is, no doubt, contemplating the implications of enantiomerism in tartaric acid crystals.

The diastereomers of 2-butyl tartrate have different physical properties, and they can be separated by conventional distillation, recrystallization, or chromatography. Separation of the diastereomers leaves us with two flasks, each containing one of the diastereomeric esters. The resolving agent is then cleaved from the separated

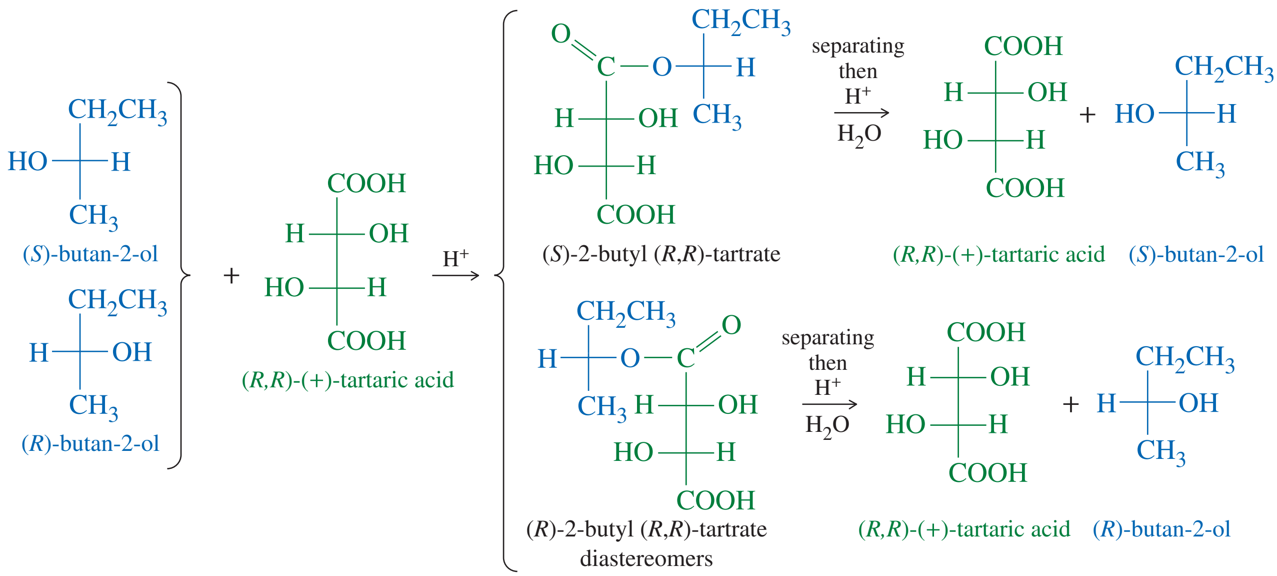
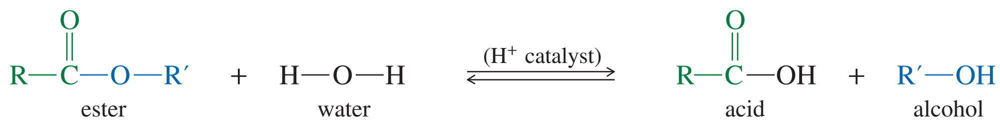


FIGURE 5-22 Formation of (*R*)- and (*S*)-2-butyl tartrate. The reaction of a pure enantiomer of one compound with a racemic mixture of another compound produces a mixture of diastereomers. Separation of the diastereomers, followed by hydrolysis, gives the resolved enantiomers.

enantiomers of butan-2-ol by the reverse of the reaction used to make the ester. Adding an acid catalyst and an excess of water to an ester drives the equilibrium toward the acid and the alcohol:



Hydrolysis of (*R*)-butan-2-ol tartrate gives (*R*)-butan-2-ol and (+)-tartaric acid, and hydrolysis of (*S*)-butan-2-ol tartrate gives (*S*)-butan-2-ol and (+)-tartaric acid. The recovered tartaric acid would probably be thrown away, because it is cheap and nontoxic. Many other chiral resolving agents are expensive, so they must be carefully recovered and recycled.

Louis Pasteur (1822–1895) was a French chemist who pioneered the field of vaccination and microbial fermentation. He was renowned for his work on pasteurization, which led to the prevention of many diseases. In recognition of Pasteur's legacy in the scientific field, the UNESCO/Institut Pasteur Medal was created on the centenary of his death.

PROBLEM 5-24

To show that (*R*)-2-butyl (*R*,*R*)-tartrate and (*S*)-2-butyl (*R*,*R*)-tartrate are not enantiomers, draw and name the mirror images of these compounds.

5-16B Chromatographic Resolution of Enantiomers

Chromatography is a powerful method for separating compounds. One type of chromatography involves passing a solution through a column containing particles whose surface tends to adsorb organic compounds. Compounds that are adsorbed strongly spend more time on the stationary particles; they come off the column later than less strongly adsorbed compounds, which spend more time in the mobile solvent phase.

In some cases, enantiomers may be resolved by passing the racemic mixture through a column containing particles whose surface is coated with chiral molecules (Figure 5-23). As the solution passes through the column, the enantiomers form weak complexes, usually through hydrogen bonding, with the chiral column packing.

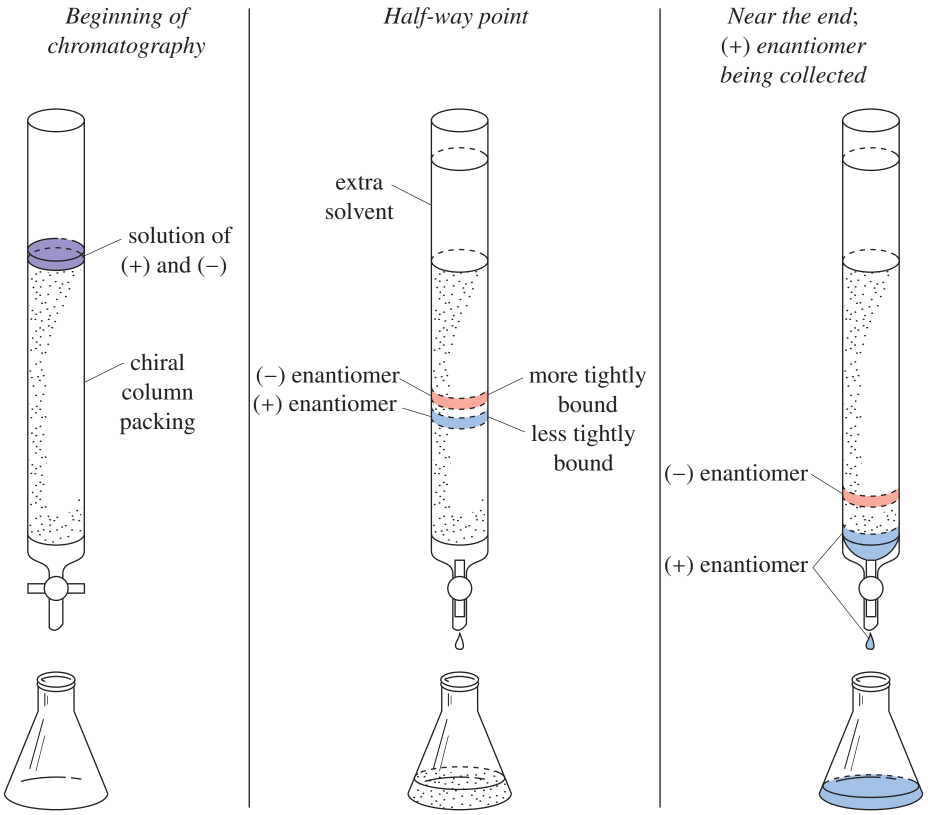


FIGURE 5-23 Chromatographic resolution of enantiomers. The enantiomers of the racemic compound form diastereomeric complexes with the chiral material on the column packing. One of the enantiomers binds more tightly than the other, so it moves more slowly through the column.

Application: Biochemistry

Enzymes can also be used to eliminate an undesired stereoisomer. The enzyme will process only one isomer in a racemic mixture and leave the other stereoisomer untouched.

The solvent flows continually through the column, and the dissolved enantiomers gradually move along, retarded by the time they spend complexed with the column packing.

The special feature of this chromatography is the fact that the enantiomers form diastereomeric complexes with the chiral column packing. These diastereomeric complexes have different physical properties. They also have different binding energies and different equilibrium constants for complexation. One of the two enantiomers will spend more time complexed with the chiral column packing. The more strongly complexed enantiomer passes through the column more slowly and emerges from the column after the faster-moving (more weakly complexed) enantiomer.

Essential Terms

absolute configuration The detailed stereochemical picture of a molecule, including how the atoms are arranged in space. Alternatively, the (*R*) or (*S*) configuration at each asymmetric carbon atom. (p. 280)

achiral Not chiral. (p. 245)

allenes Compounds having two C═C double bonds that meet at a single carbon atom, C═C═C. The two outer carbon atoms are trigonal planar, with their planes perpendicular to each other. Many substituted allenes are chiral. (p. 268)

asymmetric carbon atom **(chiral carbon atom)** A carbon atom that is bonded to four different groups. (p. 247)

Cahn–Ingold–Prelog convention The accepted method for designating the absolute configuration of a chiral center (usually an asymmetric carbon) as either (*R*) or (*S*). (p. 252)

chiral Different from its mirror image. (p. 245)

chiral carbon atom **(asymmetric carbon atom)** A carbon atom that is bonded to four different groups. (p. 247)

chiral center **(chirality center)** The IUPAC term for an atom holding a set of ligands in a spatial arrangement that is not superimposable on its mirror image. Asymmetric carbon atoms are the most common chiral centers. (p. 248)

chiral probe	A molecule or an object that is chiral and can use its own chirality to differentiate between mirror images. (p. 261)
cis	On the same side of a ring or double bond. (p. 274)
cis-trans isomers	**(geometric isomers)** Isomers that differ in their geometric arrangement on a ring or double bond; cis-trans isomers are a subclass of diastereomers. (p. 274)
configurational isomers	(see **stereoisomers**)
configurations	The two possible spatial arrangements around a chiral center or other stereocenter. (p. 252)
conformations	Structures that differ only by rotations about single bonds. In most cases, conformations interconvert at room temperature; thus, they are not different compounds and not true isomers. (p. 265)
constitutional isomers	**(structural isomers)** Isomers that differ in the order in which their atoms are bonded together. (p. 244)
D–L configurations	**(Fischer–Rosanoff convention)** D has the same relative configuration as (+)-glyceraldehyde. L has the same relative configuration as (−)-glyceraldehyde. (p. 281)
dextrorotatory, (+), or (*d*)	Rotating the plane of polarized light clockwise. (p. 258)
diastereomers	Stereoisomers that are not mirror images. (p. 274)
enantiomeric excess (e.e.)	The excess of one enantiomer in a mixture of enantiomers expressed as a percentage of the mixture. Similar to optical purity. (p. 263) Algebraically, $$e.e. = \frac{\lvert R - S \rvert}{R + S} \times 100\%$$
enantiomers	A pair of nonsuperimposable mirror-image molecules: mirror-image isomers. (p. 264)
Fischer projection	A method for drawing an asymmetric carbon atom as a cross. The carbon chain is kept along the vertical, with the IUPAC numbering from top to bottom. Vertical bonds project away from the viewer, and horizontal bonds project toward the viewer. (p. 269)
geometric isomers	(see **cis-trans isomers**) (p. 274)
internal mirror plane (σ)	A plane of symmetry through the middle of a molecule, dividing the molecule into two mirror-image halves. A molecule with an internal mirror plane of symmetry cannot be chiral. (p. 250)
isomers	Different compounds with the same molecular formula. (p. 276)
Leftorium	A store that sells the enantiomers of everyday chiral objects such as scissors, rifles, can openers, etc. (p. 245)
levorotatory, (−), or (*l*)	Rotating the plane of polarized light counterclockwise. (p. 259)
meso compound	An achiral compound that contains chiral centers (usually asymmetric carbon atoms). Originally, an achiral compound that has chiral diastereomers. (p. 277)
optical activity	Rotation of the plane of polarized light. (p. 257)
optically active	Capable of rotating the plane of polarized light. (p. 257)
optical isomers	**(archaic; see enantiomers)** Compounds with identical properties except for the direction in which they rotate polarized light. (p. 258)
optical purity (o.p.)	The specific rotation of a mixture of two enantiomers, expressed as a percentage of the specific rotation of one of the pure enantiomers. Similar to enantiomeric excess. (p. 263) Algebraically, $$\text{o.p.} = \frac{\text{observed rotation}}{\text{rotation of pure enantiomer}} \times 100\%$$
plane-polarized light	Light composed of waves that vibrate in only one plane. (p. 256)
polarimeter	An instrument that measures the rotation of plane-polarized light by an optically active compound. (p. 258)
racemic mixture	**[racemate, racemic modification, (±) pair, (*d,l*) pair]** A mixture of equal quantities of enantiomers, such that the mixture is optically inactive. (p. 262)
relative configuration	The experimentally determined relationship between the configurations of two molecules, even though the absolute configuration of either may not be known. (p. 281)
resolution	The process of separating a racemic mixture into the pure enantiomers. Resolution requires a chiral resolving agent. (p. 284)
resolving agent	A chiral compound (or chiral material on a chromatographic column) used for separating enantiomers. (p. 284)

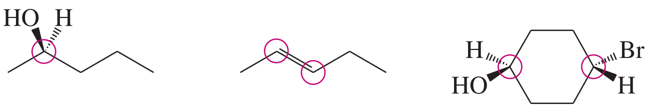
2^n rule A molecule with n chiral centers might have as many as 2^n stereoisomers. (p. 277)

specific rotation A measure of a compound's ability to rotate the plane of polarized light, given by

$$[\alpha]_D^{25} = \frac{\alpha(\text{observed})}{c \cdot l}$$

where c is concentration in g/mL and l is length of sample cell (path length) in decimeters. (p. 259)

stereocenter **(stereogenic atom)** An atom that gives rise to stereoisomers when its groups are interchanged. Asymmetric carbon atoms and double-bonded carbons in cis-trans alkenes are the most common stereocenters. (p. 248)



examples of stereocenters (circled)

stereochemistry The study of the three-dimensional structure of molecules. (p. 244)

stereogenic Giving rise to stereoisomers. Characteristic of an atom or a group of atoms such that interchanging any two groups creates a stereoisomer. (p. 248)

stereoisomers **(configurational isomers)** Isomers whose atoms are bonded together in the same order but differ in how the atoms are oriented in space. (p. 244)

structural isomers (see **constitutional isomers**) Isomers that differ in the order in which their atoms are bonded together. (p. 244)

superimposable **(congruent)** Identical in all respects. The three-dimensional positions of all atoms coincide when the molecules are placed on top of each other. (p. 246)

trans On opposite sides of a ring or double bond. (p. 274)

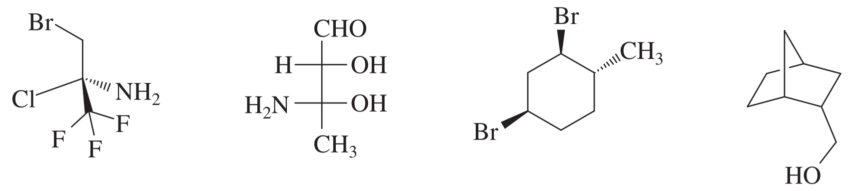
Essential Problem-Solving Skills in Chapter 5

Each skill is followed by problem numbers exemplifying that particular skill.

1	Draw all the stereoisomers of a given structure, and identify the relationships between the stereoisomers.	Problems 5-27, 30, 35, and 37
2	Classify molecules as chiral or achiral, and draw their mirror images. Identify and draw any mirror planes of symmetry.	Problems 5-26, 27, 31, 35, and 37
3	Identify asymmetric carbon atoms and other chiral centers and name them using the (*R*) and (*S*) nomenclature.	Problems 5-25, 26, 27, 36, 37, and 38
4	Calculate specific rotations from polarimetry data, and use specific rotations to determine the optical purity and the enantiomeric excess of mixtures.	Problems 5-32, 33, and 34
5	Use Fischer projections to represent the stereochemistry of compounds with one or more asymmetric carbon atoms.	Problems 5-28, 29, 30, and 37
6	Identify pairs of enantiomers, diastereomers, and meso compounds, and explain how they differ in their physical and chemical properties.	Problems 5-30, 31, and 36
7	Explain how different types of stereoisomers can be separated.	Problems 5-30 and 36

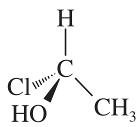
Study Problems

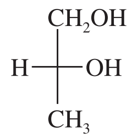
5-25 The following four structures are optically active compounds. Star (*) the asymmetric carbon atoms in these structures.

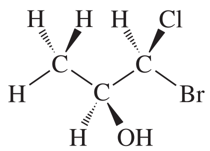


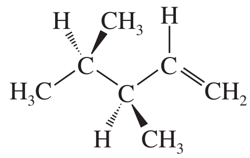
5-26 For each structure,

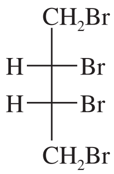
1. star (*) any asymmetric carbon atoms.
2. label each asymmetric carbon as (*R*) or (*S*).
3. draw any internal mirror planes of symmetry.
4. label the structure as chiral or achiral.
5. label any meso structures.

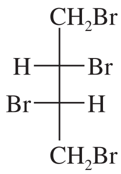
(a) 

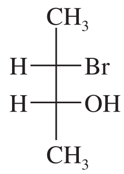
(b) 

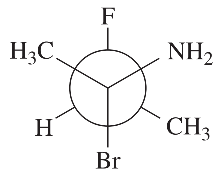
(c) 

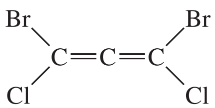
(d) 

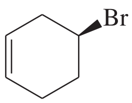
(e) 

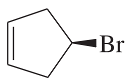
(f) 

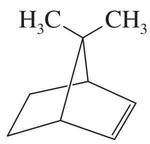
(g) 

(h) 

(i) 

(j) 

(k) 

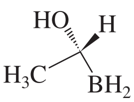
(l) 

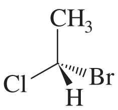
5-27 For each of the compounds described by the following names,

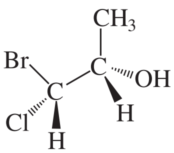
1. draw a three-dimensional representation.
2. star (*) each chiral center.
3. draw any planes of symmetry.
4. draw any enantiomer.
5. draw any diastereomers.
6. label each structure you have drawn as chiral or achiral.

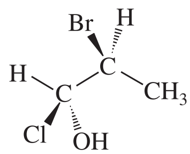
(a) (*S*)-2-chlorobutane
(b) (*R*)-1,1,2-trimethylcyclohexane
(c) (2*R*,3*S*)-2,3-dibromohexane
(d) (1*R*,2*R*)-1,2-dibromocyclohexane
(e) *meso*-hexane-3,4-diol, $CH_3CH_2CH(OH)CH(OH)CH_2CH_3$
(f) (±)-hexane-3,4-diol

5-28 Convert the following perspective formulas to Fischer projections.

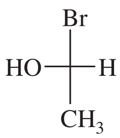
(a) 

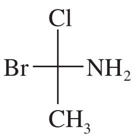
(b) 

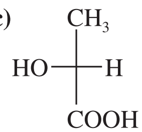
(c) 

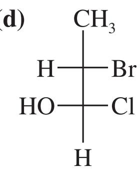
(d) 

5-29 Convert the following Fischer projections to perspective formulas.

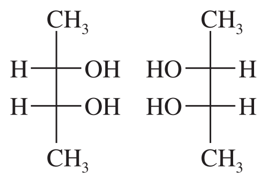
(a) 

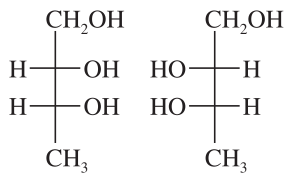
(b) 

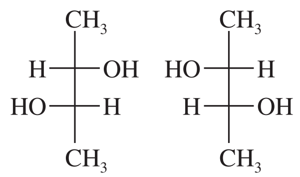
(c) 

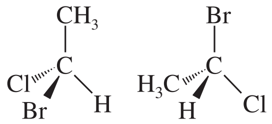
(d) 

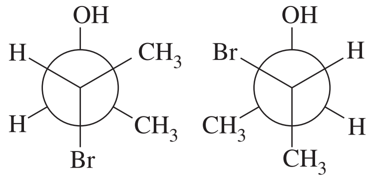
5-30 Give the stereochemical relationships between each pair of structures. Examples are same compound, structural isomers, enantiomers, diastereomers. Which pairs could you (theoretically) separate by distillation or recrystallization?

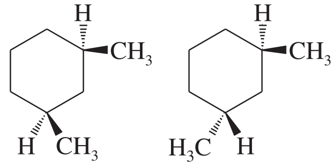
(a) 

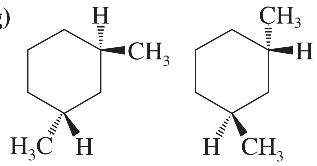
(b) 

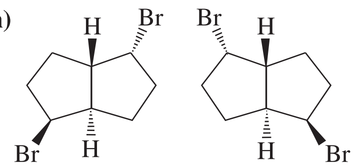
(c) 

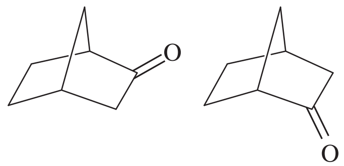
(d) 

(e) 

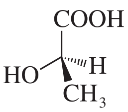
(f) 

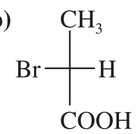
(g) 

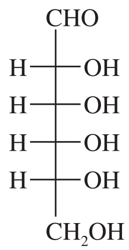
***(h)** 

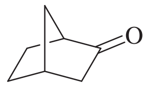
(i) 

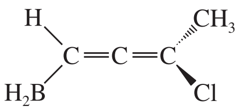
5-31 Draw the enantiomer, if any, for each structure.

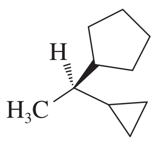
(a) 

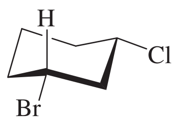
(b) 

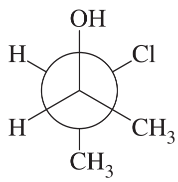
(c) 

(d) 

(e) 

(f) 

(g) 

(h) 

5-32 Calculate the specific rotations of the following samples taken at 25 °C using the sodium D line.

(a) 1.00 g of sample is dissolved in 20.0 mL of ethanol. Then 5.00 mL of this solution is placed in a 20.0-cm polarimeter tube. The observed rotation is 2.25° counterclockwise.

(b) 0.050 g of sample is dissolved in 2.0 mL of ethanol, and this solution is placed in a 2.0-cm polarimeter tube. The observed rotation is clockwise 0.058°.

5-33 (+)-Tartaric acid has a specific rotation of +12.0°. Calculate the specific rotation of a mixture of 68% (+)-tartaric acid and 32% (−)-tartaric acid.

5-34 The specific rotation of (*S*)-2-iodobutane is +15.90°.

(a) Draw the structure of (*S*)-2-iodobutane.

(b) Predict the specific rotation of (*R*)-2-iodobutane.

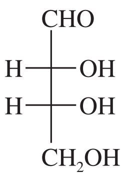
(c) Determine the percentage composition of a mixture of (*R*)- and (*S*)-2-iodobutane with a specific rotation of −7.95°.

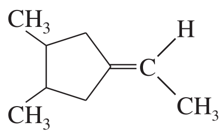
5-35 For each structure,

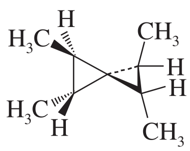
1. draw all the stereoisomers.

2. label each structure as chiral or achiral.

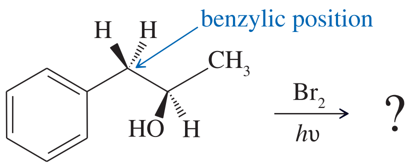
3. give the relationships between the stereoisomers (enantiomers, diastereomers).

(a) 

***(b)** 

****(c)** 

*5-36 Free-radical bromination of the following compound introduces bromine primarily at the benzylic position next to the aromatic ring. If the reaction stops at the monobromination stage, two stereoisomers result.

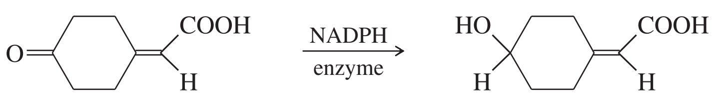


(a) Propose a mechanism to show why free-radical halogenation occurs almost exclusively at the benzylic position.
(b) Draw the two stereoisomers that result from monobromination at the benzylic position.
(c) Assign *R* and *S* configurations to the asymmetric carbon atoms in the products.
(d) What is the relationship between the two isomeric products?
(e) Will these two products be produced in identical amounts? That is, will the product mixture be exactly 50:50?
(f) Will these two stereoisomers have identical physical properties such as boiling point, melting point, solubility, etc.? Could they be separated (theoretically, at least) by distillation or recrystallization?

*5-37 If you think you know your definitions, try this difficult problem.
(a) Draw all the stereoisomers of 2,3,4-tribromopentane. (Using Fischer projections may be helpful.) You should find two meso structures and one pair of enantiomers.
(b) Star (*) the asymmetric carbon atoms, and label each as (*R*) or (*S*).
(c) In the meso structures, show how C3 is not asymmetric, nor is it a chiral center, yet it is a stereocenter.
(d) In the enantiomers, show how C3 is not a stereocenter.

*5-38 3,4-Dimethylpent-1-ene has the formula $CH_2{=}CH{-}CH(CH_3){-}CH(CH_3)_2$. When pure (*R*)-3,4-dimethylpent-1-ene is treated with hydrogen over a platinum catalyst, the product is (*S*)-2,3-dimethylpentane.
(a) Draw the equation for this reaction. Show the stereochemistry of the reactant and the product.
(b) Has the chiral center retained its configuration during this hydrogenation, or has it been inverted?
(c) The reactant is named (*R*), but the product is named (*S*). Does this name change imply a change in the spatial arrangement of the groups around the chiral center? So why does the name switch from (*R*) to (*S*)?
(d) How useful is the (*R*) or (*S*) designation for predicting the sign of an optical rotation? Can you predict the sign of the rotation of the reactant? Of the product? (*Hint* from Juliet Capulet: "What's in a name? That which we call a rose/By any other name would smell as sweet.")

*5-39 A graduate student was studying enzymatic reductions of cyclohexanones when she encountered some interesting chemistry. When she used an enzyme and NADPH to reduce the following ketone, she was surprised to find that the product was optically active. She carefully repurified the product so that no enzyme, NADPH, or other contaminants were present. Still, the product was optically active.



optically active

(a) Does the product have any asymmetric carbon atoms or other stereocenters?
(b) Is the product capable of showing optical activity? If it is, explain how.
(c) If this reaction could be accomplished using H_2 and a nickel catalyst, would the product be optically active? Explain.

*5-40 D-(−)-Erythrose has the formula $HOCH_2{-}CH(OH){-}CH(OH){-}CHO$, and the D in its name implies that it can be degraded to D-(+)-glyceraldehyde. The (−) in its name implies that D-(−)-erythrose is optically active (levorotatory). When D-(−)-erythrose is reduced (using H_2 and a nickel catalyst), it gives an optically inactive product of formula $HOCH_2{-}CH(OH){-}CH(OH){-}CH_2OH$. Knowing the absolute configuration of D-(+)-glyceraldehyde (Section **5-**14), determine the absolute configuration of D-(−)-erythrose.

*5-**41** The original definition of *meso* is "an achiral compound that has chiral diastereomers." Our working definition of *meso* is "an achiral compound that has chiral centers (usually asymmetric carbon atoms)." The working definition is much easier to apply, because we don't have to envision all possible chiral diastereomers of the compound. Still, the working definition is not quite as complete as the original definition.
(a) Show how *cis*-cyclooctene is defined as a *meso* compound under the original definition, but not under our working definition. (Review Figure 5-19.)
(b) See if you can construct a double allene that is achiral, although it has chiral diastereomers, and is therefore a *meso* compound under the original definition. The allene structure is not a chiral center, but it can be a *chiral axis*.